

A binomial count of meromorphic travelling waves

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Abstract

We study meromorphic travelling waves of $u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0$, a constant-coefficient family containing KdV–Burgers, Kuramoto–Sivashinsky and Kawahara. After integration and normalization, the profiles satisfy $P(D)v + v^2/2 = 0$, where $D = d/dz$ and P is monic of degree $p \geq 2$ with complex coefficients.

For evolution equations of even order (p odd), and for purely dispersive equations (only odd-order derivatives), we give a complete algebraic description of all nonconstant globally meromorphic travelling waves: they are rational, rational in an exponential, or elliptic, with one pole per period.

One-pole waves of these three kinds are exceptional: for $p \geq 3$ a generic equation of the family has none. We count the exceptions as pairs (P, v) , over the whole family. Rescaling x changes P and the period of v together, so the period is held fixed. There are $\binom{2p-1}{p-1}$ such pairs with v rational in an exponential of given rate, and $2\binom{2p-2}{p-2}$ with v elliptic on a given generic lattice. We also count how many of these belong to the purely dispersive subfamily (p even) and to the purely dissipative one (only even-order derivatives in the linear part, p odd). The subfamily present at a given p contains $F_p = \binom{p-1}{\lfloor (p-1)/2 \rfloor}$ of the rational-exponential pairs: all solitary pulses in the dispersive case, all fronts in the dissipative case. The purely dispersive subfamily also contains $2F_p$ of the elliptic pairs.

When $2p - 1$ is prime, these statements are proved and all pairs are distinct. Exact computer certificates establish the counts, with multiplicities, for $2 \leq p \leq 73$. We conjecture the same in every order.

Keywords. Meromorphic travelling waves; dispersive and dissipative equations; elliptic functions; polynomial convolution; algebraic multiplicity.

1 Introduction

Exact travelling waves of nonlinear evolution equations are usually found one equation at a time, and usually only for special values of the coefficients. The generalized Kuramoto–Sivashinsky equation is a classical example. It has nonconstant travelling waves that are polynomials in a hyperbolic tangent at exactly four values of its squared dimensionless dispersion parameter. Kuramoto and Tsuzuki found the dispersionless one [39], Kudryashov the other three [26, 27], and Eremenko proved that, together with Kudryashov’s elliptic solution and a single rational one, they exhaust the nonconstant meromorphic travelling waves [19]. The classical tables list six cases at these four values [5, 33]. We count a wave and its mirror image separately. The two waves of the dispersionless equation are their own mirror images and the other four are not, so the list has ten members. This paper shows that ten is $\binom{5}{2}$: the classical list is the case $p = 3$ of a binomial enumeration, proved for infinitely many p and conjectured for all.

We consider the evolution equations

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0,$$

which contain KdV–Burgers ($p = 2$), Kuramoto–Sivashinsky ($p = 3$) and Kawahara ($p = 4$), the sixth-order turbulence models of Kudryashov and Migita [32], Simbawa et al. [46], and the quadratic long-wave reductions of Kudryashov et al. [34]. A travelling wave $u = w(x - ct)$ satisfies, after one integration, $\sum_j b_j w^{(j)} - cw + w^2/2 + K = 0$. Shifting w by a constant solution and normalizing the leading coefficient gives

$$E_P(v) := P(D)v + \frac{1}{2}v^2 = 0, \quad P(\rho) = \rho^p + \sum_{j=0}^{p-1} a_j \rho^j, \quad p \geq 2, \quad (1.1)$$

with $D = d/dz$ and complex coefficients. The speed, the integration constant and the background are absorbed into a_0 (Section 2).

Throughout, p is the order of the profile equation (1.1). The evolution equation has order $p + 1$: Kuramoto–Sivashinsky is a fourth-order equation and has $p = 3$, and Kawahara is a fifth-order equation and has $p = 4$. Phrases such as “order three” or “in every order” always refer to p . The order of the evolution equation appears only in customary names, such as the seventh-order KdV equation, which has $p = 6$. We call $2p - 1$ the *index* of the order p .

Two subfamilies play a special role. In the evolution equation, derivatives of odd order are dispersive terms and derivatives of even order are dissipative terms. We call an equation *purely dispersive* if it contains only odd-order derivatives, as the KdV and Kawahara equations do; then p is even and P is an even polynomial. We call it *purely dissipative* if its linear part contains only even-order derivatives, as for the Kuramoto–Sivashinsky equation without dispersion; then p is odd and $P - P(0)$ is an odd polynomial. All other equations, such as KdV–Burgers, are *mixed*. Both pure subfamilies have a reflection symmetry, which is described in Section 2.

Classification. Eremenko’s theorem states that an autonomous algebraic differential equation with a single monomial of top degree and finitely many Laurent expansions at a pole has only three kinds of nonconstant meromorphic solutions: rational, rational in an exponential, and elliptic [19]. Theorem 3.1 verifies these hypotheses for (1.1) in two cases. The first is p odd, that is, every evolution equation of even order: there the Laurent expansion at a pole is unique. The second is the purely dispersive case: there one Laurent coefficient is free, but (1.1) is invariant under $z \mapsto -z$ and conserves an energy, and fixing the energy fixes that coefficient. In both cases every nonconstant meromorphic travelling wave is of one of the three kinds and has one pole per period. For a fixed operator P there is at most one such profile up to translation when p is odd, and at most one on each energy level when the equation is purely dispersive. Mixed operators with p even, such as that of KdV–Burgers, are not covered, and cannot be: Section 10 recalls an example with $p = 2$ that has other meromorphic waves. Section 4 writes each of the three kinds as a finite polynomial system, following the principal-part construction of Demina and Kudryashov [16, 17].

The count. One-pole waves of these three kinds are exceptional. In every order $p \geq 3$ for which the count below is established, a generic equation of the family has none of them (Section 8). Here and below, generic means outside a proper algebraic subset of the coefficients, or of the lattices. It refers to the full family: a pure subfamily of low order, such as the KdV or Kawahara equations, can lie on the exceptional set. We count the exceptions.

Translations and changes of length scale produce continuous families of equivalent descriptions, so before asking for a finite count we fix these freedoms. For a rational-exponential wave we prescribe its primitive period, or equivalently its exponential rate. For an elliptic wave we prescribe its full period lattice, including its scale. We then place the pole at a fixed point. The

coefficients of P still vary: we count which normalized equations admit such a wave, together with the wave itself. Precisely, a rational-exponential pair (P, v) has v rational in e^z , with one pole modulo $2\pi i\mathbb{Z}$, placed at $i\pi$, and $v \rightarrow 0$ as $e^z \rightarrow 0$. An elliptic pair has v elliptic on a fixed nonsingular lattice, with its only pole at zero; each elliptic wave is then counted on both of its constant backgrounds, which is the factor two below. In both cases P ranges over all monic polynomials of degree p . The answers are

$$\begin{aligned} N_p &= \binom{2p-1}{p-1} && \text{rational-exponential pairs,} \\ M_p &= 2 \binom{2p-2}{p-2} && \text{elliptic pairs on a generic lattice.} \end{aligned}$$

The pairs are the solutions of a polynomial system with as many equations as unknowns, and a solution can be simple or multiple, like a root of a polynomial. Whenever $2p-1$ is prime, both formulas hold and every pair has multiplicity one, so there are exactly N_p distinct pairs, and exactly M_p on a generic lattice (Theorems 5.4 and 6.1). For every $2 \leq p \leq 73$, the formulas hold when each pair is counted with its multiplicity (Section 7); the number of distinct pairs is then at most N_p or M_p , with equality when no pair is multiple. On special lattices elliptic pairs do merge, and Section 9.2 determines where this happens at order three. At $p=5$, where $2p-1=9$ is the first composite index, a separate exact computation shows that all 126 rational-exponential pairs have multiplicity one. We conjecture the same counting and multiplicity statements in every order (Conjecture 10.1). The counts are over $\mathbb{C}$ and range over all monic P , mixed operators included. The classification is used only to know that, where it applies, there are no other meromorphic waves. Rational waves are not counted: under the rigidity condition of Section 5 the only one is $v = c_* z^{-p}$, for $P = D^p$.

Purely dispersive and purely dissipative equations. Reading a wave in the opposite direction, $z \mapsto -z$, and restoring the normalization sends pairs to pairs. The pairs that coincide with their mirror image are exactly those whose equation is purely dispersive, for even p , or purely dissipative, for odd p (Theorem 5.8). Their number is

$$F_p = \binom{p-1}{\lfloor (p-1)/2 \rfloor},$$

with the same status as N_p : it is proved, with all pairs distinct, when $2p-1$ is prime, and as a count with multiplicity for $2 \leq p \leq 73$. The purely dispersive equations also account for $2F_p$ of the elliptic pairs on a generic lattice (Corollary 6.2).

The two subfamilies carry different kinds of rational-exponential waves. Those of a purely dispersive equation are all pulses, which return to the same constant at both ends; this holds for every even p . Those of a purely dissipative equation are all fronts, which connect two different constants. This second statement is proved for every p at which all N_p pairs are distinct, in particular when $2p-1$ is prime, and for waves with real coefficients at every odd p . For example, the Kawahara family has three rational-exponential pairs and six elliptic pairs on a generic lattice, which are F_4 and $2F_4$; the seventh-order KdV family has ten pairs, two of them real; and the six pairs of the purely dissipative family with $p=5$ contain the four kink solutions of Kudryashov and Migita [32] (Section 9).

Table 1 lists the resulting counts for the named equations of the family with $p \leq 6$: the Benney–Lin equation [2, 40], the Kuramoto–Sivashinsky equation with dispersion, also called

the Benney or KdV–KS equation [25], the Nikolaevskiy equation [44] and its dispersive version [32, 46], and the equations already mentioned.

Table 1: The named equations of the family with $p \leq 6$; here $u_{kx} = \partial_x^k u$. Each name stands for a family with free coefficients, and the numbers count normalized pairs across that family: rational-exponential pairs at unit rate, and elliptic pairs on a generic fixed lattice. A full family has the counts N_p and M_p , and it contains the pure subfamily of the same p : the three pairs of KdV–Burgers include the KdV soliton. A pure subfamily has F_p rational-exponential pairs, and $2F_p$ elliptic pairs or none. All numbers in this table are counts of distinct pairs.

Equation	Linear terms	p	Family	Rat.-exp.	Elliptic
KdV	u_{3x}	2	purely dispersive	1	2
KdV–Burgers	u_{2x}, u_{3x}	2	full	3	2
Kuramoto–Sivashinsky (KS)	u_{2x}, u_{4x}	3	purely dissipative	2	0
KS with dispersion	u_{2x}, u_{3x}, u_{4x}	3	full	10	8
Kawahara	u_{3x}, u_{5x}	4	purely dispersive	3	6
Benney–Lin	$u_{2x}, \dots, u_{5x}$	4	full	35	30
Nikolaevskiy	u_{2x}, u_{4x}, u_{6x}	5	purely dissipative	6	0
Nikolaevskiy with dispersion	$u_{2x}, \dots, u_{6x}$	5	full	126	112
seventh-order KdV	u_{3x}, u_{5x}, u_{7x}	6	purely dispersive	10	20

A single equation. A count across the family is not a count for each individual equation. To recover the waves of a given equation, we restore the length scale and retain only the normalized pairs whose coefficients match it. Changing the scale traces a curve in the space of coefficients. Thus, in the orders covered by the count, the equations with a one-pole rational-exponential wave lie on finitely many such scaling curves. If at least one lower-derivative coefficient is nonzero, only finitely many wave shapes remain after translations and Galilean shifts are identified. The scale-invariant case is different: the KdV equation, for example, retains its family of solitons of different widths. Section 8 gives the precise criterion, recovers the four familiar Kuramoto–Sivashinsky coefficient curves, and shows that for odd $p \geq 3$ in the settled range a generic equation has no nonconstant meromorphic travelling wave at all.

Method. Every rational-exponential pair has a profile of the form $v_A = s_p A(D)Q$, where $Q = 1/(1 + e^{-z})$, s_p is a fixed scale and A is a monic polynomial of degree $p - 1$. The pair solves (1.1) if and only if A divides its own discrete self-convolution $S_A(n) = \sum_{j=1}^{n-1} A(j)A(n-j)$ (Proposition 5.1). When $\ell = 2p - 1$ is prime, this condition becomes $A \mid \rho^\ell - \rho$ modulo ℓ , so its solutions are labelled by the $(p - 1)$ -element subsets S of $\mathbb{F}_\ell$. This is where the binomial coefficient comes from. Each label lifts to exactly one solution, and the reflection acts on the labels by $S \mapsto -S$, which gives F_p . The elliptic count uses two limits of the lattice. When both periods become infinite, the elliptic problem becomes a rational one, and this limit gives the count with multiplicity on every nonsingular lattice. When one period becomes infinite, it becomes the pulse case of the exponential problem, with two choices of background, and this limit shows that the pairs have multiplicity one on a generic lattice.

Two ingredients are not yet proved in every order. The first is a rigidity statement: the only rational solution of (1.1) with one pole that vanishes at infinity is $c_* z^{-p}$, with $P = D^p$.

Equivalently, a monic polynomial A of degree $p - 1$ that divides its continuous self-convolution $\int_0^p A(t)A(\rho - t) dt$ is ρ^{p-1} . Rigidity gives all counts with multiplicity. It is proved when $2p - 1$ is a prime or a prime power, and certified by exact arithmetic for $p \leq 73$. The second is that every rational-exponential pair has multiplicity one, which makes the pairs distinct. It is proved when $2p - 1$ is prime and, by exact computation, at $p = 5$. Both are conjectured in general (Conjecture 10.1).

Relation to earlier work. The classification theorem is a uniform version of results that exist in pieces. The case of odd p is contained in Yuan et al. [49, Theorem 1], and the energy device for even P is that of Eremenko et al. [20, Lemmas 3–4], which has been applied to the Kawahara, generalized Bretherton and seventh-order KdV equations; Section 3 gives the detailed attributions. The polynomial systems are instances of the principal-part construction of Demina and Kudryashov [16, 17]. The general enumeration has no precedent that we know of; its special cases at $p = 3$ and in the Kawahara family are classical, as the examples make explicit.

Organization. Section 2 fixes the normalizations and the vocabulary. Sections 3 and 4 contain the classification and the three polynomial systems. Sections 5 and 6 prove the two counts, Section 7 records which orders are settled, and Section 8 reads the count for a single equation. Section 9 works out the lowest orders and the two pure subfamilies, including the lattices on which elliptic pairs merge. Section 10 collects the open questions, and Appendix A describes the exact verification and the ancillary material. All counts are over $\mathbb{C}$; the examples identify the real profiles, and stability is not discussed.

2 Equations, waves and what is counted

This section fixes the normalizations and the vocabulary used in the rest of the paper. The normalizations are what makes a finite count possible, and they are also what connects the count with a given evolution equation.

2.1 From the evolution equation to the profile equation

Let $u = w(\xi)$, $\xi = x - ct$, be a travelling wave of

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0. \quad (2.1)$$

One integration gives

$$\sum_{j=1}^p b_j w^{(j)} - cw + \frac{1}{2}w^2 + K = 0 \quad (2.2)$$

with a constant K . Let w_* be a constant solution of (2.2), that is, a root of $w^2/2 - cw + K$, and let $\lambda \neq 0$. The substitution

$$w(\xi) = w_* + b_p \lambda^p v(\lambda \xi) \quad (2.3)$$

turns (2.2) into the profile equation (1.1) with

$$a_j = \frac{b_j}{b_p} \lambda^{j-p} \quad (1 \leq j \leq p-1), \quad a_0 = \frac{w_* - c}{b_p \lambda^p}. \quad (2.4)$$

Thus $a_1, \dots, a_{p-1}$ record the equation, up to the scale λ , and a_0 records the speed of the wave relative to the background w_* . The scale is free at this stage. It will be fixed by the period of the wave.

The constant solutions of (1.1) are $v = 0$ and $v = -2a_0$. They correspond to the two possible choices of the background w_* . Changing that choice replaces the pair (P, v) by

$$(P - 2P(0), v + 2P(0)), \quad (2.5)$$

which again satisfies (1.1), as a direct substitution shows. After the original background is restored, the two pairs describe the same wave of the same evolution equation. They coincide exactly when $a_0 = 0$.

2.2 Dispersive, dissipative and mixed equations

In (2.1) a derivative of odd order is a dispersive term, and a derivative of even order is a dissipative term: for real coefficients the first changes the phase of a Fourier mode and the second its amplitude. We use the following names.

- The equation is *purely dispersive* if it contains only odd-order derivatives, that is, $b_j = 0$ for odd j . Then p is even, P is an even polynomial, and the equation is invariant under $(x, t) \mapsto (-x, -t)$. Examples are the KdV equation ($p = 2$), the Kawahara equation ($p = 4$) and the seventh-order KdV equation ($p = 6$).
- The equation is *purely dissipative* if its linear part contains only even-order derivatives, that is, $b_j = 0$ for even j . Then p is odd, $P - P(0)$ is an odd polynomial, and the equation is invariant under $(x, u) \mapsto (-x, -u)$. Examples are the Kuramoto–Sivashinsky equation without dispersion ($p = 3$) and the Nikolaevskiy equation ($p = 5$); see Simbawa et al. [46].
- All other equations are *mixed*. Examples are the KdV–Burgers equation and the Kuramoto–Sivashinsky equation with dispersion.

The two pure subfamilies are described by one condition. We call a monic polynomial P of degree p *reflection-symmetric* if $P - P(0)$ has the parity of ρ^p :

$$P(-\rho) = P(\rho) \quad (p \text{ even}), \quad P(\rho) + P(-\rho) = 2P(0) \quad (p \text{ odd}). \quad (2.6)$$

In the travelling-wave variable both symmetries are the reflection $z \mapsto -z$. If P is even and $v(z)$ solves (1.1), so does $v(-z)$. If p is odd and $P - P(0)$ is odd, so does $-v(-z) - 2P(0)$; this is the reflection combined with the change of background (2.5).

2.3 Waves, poles, pulses and fronts

Throughout, a meromorphic function is single-valued and meromorphic on the whole complex plane. Three kinds of meromorphic travelling waves occur in this paper: rational in z , rational in an exponential $e^{\kappa z}$, and elliptic. For the equations covered by Theorem 3.1 there are no others; for the remaining equations these are the waves that we count. The period group of such a wave is its *full* group of periods: trivial in the first case, infinite cyclic in the second, a lattice in the third. We say that the wave has *one pole per period* if its poles form a single coset of this group. There is then one pole in the rational case, one pole modulo the primitive period in the second case, and one pole in each period parallelogram in the elliptic case.

A wave that is rational in $e^{\kappa z}$ and has one pole per period has finite limits as $e^{\kappa z} \rightarrow 0$ and ∞ (Section 4). Both limits are constant solutions. The wave is a *pulse* if the two limits coincide and a *front* otherwise. For a real rate and a real profile these are the solitary waves and the kinks of the evolution equation, and they are smooth and bounded on the real axis when the pole lies off it. In (2.3) we take w_* to be the limit as $e^{\kappa z} \rightarrow 0$. Then $v \rightarrow 0$ at that end, and the other limit is 0 for a pulse and $-2a_0$ for a front.

2.4 What is counted

We count pairs (P, v) : a monic polynomial P of degree p and a solution v of (1.1) of a given kind, with its periods fixed.

- A *rational-exponential pair* has v rational in e^z , with one pole modulo $2\pi i\mathbb{Z}$, placed at $i\pi$, and $v \rightarrow 0$ as $e^z \rightarrow 0$.
- An *elliptic pair* on a lattice Λ has v elliptic with period lattice Λ , including its scale, and its only pole modulo Λ at zero.

Fixing the period or the period lattice removes the scale λ in (2.3), and placing the pole removes translations. The operator P varies over *all* monic polynomials of degree p , and all counts are over $\mathbb{C}$. A wave and its mirror image are counted as two pairs when they are distinct; the pairs that coincide with their mirror image are the subject of Section 5.4. In the elliptic case both pairs (2.5) are counted; in the rational-exponential case the condition $v \rightarrow 0$ selects one of them. Section 8 translates the counts back to a single evolution equation.

3 The classification theorem

This section proves Theorem 3.1 below: for every evolution equation of even order (p odd), and for every purely dispersive equation (P even), a nonconstant meromorphic solution of (1.1) is rational, rational in an exponential, or elliptic, and has one pole per period. The proof has three steps. We first compute the Laurent expansion at a pole and show that for odd p it is unique. For even P it is not: one coefficient is left free, and we show that the energy, a first integral of (1.1), fixes it, so that Eremenko's theorem applies on each energy level. Finally we show that any two poles of one solution differ by a period.

Let $\mathcal{W}$ consist of rational functions of z , rational functions of $e^{\kappa z}$ with $\kappa \neq 0$, and elliptic functions. Period groups are full period groups, as in Section 2; for a rational function of $e^{\kappa z}$ we choose κ so that $2\pi i/\kappa$ generates the group. For the general analytic background, see Eremenko [18].

Theorem 3.1 below collects the analytic input to the count. It is a uniform version of results that exist in pieces. For odd p the indicial polynomial has no positive integer zero, the Laurent expansion at a pole is unique, and Eremenko's theorem applies directly; this case is contained in Yuan et al. [49, Theorem 1], with related results for odd p in Huang et al. [22], Yuan et al. [50] and a fifth-order instance in Dang [11]. For even p the Laurent recurrence has a resonance at $n = 3p$, where one coefficient is left free, and the conclusion need not hold for mixed operators: the Fisher equation has an elliptic-exponential family of solutions [1, 6, 30] whose generic members are meromorphic and of none of the three kinds. The familiar Ablowitz–Zeppetella front is a rational-exponential degeneration of this family. When P is an even polynomial the equation has a first integral, the energy, and fixing its value removes the resonant freedom. This is the

device of Eremenko et al. [20, Lemmas 3–4] for $P = D^p + a_0$, stated as a general step by Conte et al. [7, 8] and applied to the Kawahara equation [15, 48], the generalized Bretherton equation [16, 38] and the seventh-order KdV equation [10]. Theorem 3.1 extends it uniformly to every even P , intermediate derivatives included. Equations with two dominant monomials, or without the finiteness property (finitely many Laurent expansions at a pole), are treated by other methods in Demina [13, 14]. We give the proofs in full because Sections 4 and 5 use the Laurent recurrence and the energy dependence in explicit form.

Theorem 3.1 (Classification). *Assume either that p is odd or that P is an even polynomial; that is, the evolution equation has even order or is purely dispersive. Every nonconstant meromorphic solution of (1.1) belongs to $\mathcal{W}$. Every pole has order p , and the pole set is one coset of the full period group: the solution has one pole per period. Consequently there is one pole in the rational case, one pole modulo the primitive period in the simply periodic case, and one pole on the full elliptic period torus.*

For fixed P of odd degree, there is at most one nonconstant solution with its pole at zero. For even P , the same statement holds after fixing the conserved energy. The only entire solutions are $v = 0$ and $v = -2a_0$.

An even polynomial is one in which every odd power of ρ has coefficient zero, so that $P(D)$ contains only even derivatives. The hypothesis does not cover the mixed operators with p even, and it cannot: Section 10 gives a mixed second-order operator with a meromorphic solution outside $\mathcal{W}$.

3.1 Pole order and the Laurent recurrence

Write a pole expansion at zero as

$$v(z) = \sum_{n \geq 0} c_n z^{n-p}. \quad (3.1)$$

The pole order is forced by dominant balance. A pole of order r makes $P(D)v$ of order $r + p$ and v^2 of order $2r$; the two can cancel only if $r = p$. Comparing the leading coefficients gives

$$c_0 = c_* := -2(-1)^p R_p, \quad R_p = \frac{(2p-1)!}{(p-1)!}. \quad (3.2)$$

For the remaining coefficients, set

$$I_p(n) = \prod_{j=p}^{2p-1} (n-j) + c_*. \quad (3.3)$$

The product is the factor produced by D^p acting on z^{n-p} , and c_* is the factor produced by the cross term of $c_n z^{n-p}$ with the leading pole in $v^2/2$. With $c_n = 0$ for $n < 0$, comparison of the coefficients of z^{n-2p} in (1.1) gives

$$I_p(n)c_n = - \sum_{j=0}^{p-1} a_j (n-2p+j) \underline{j} c_{n-p+j} - \frac{1}{2} \sum_{k=1}^{n-1} c_k c_{n-k}, \quad n \geq 1. \quad (3.4)$$

The underline denotes a falling factorial. The right side contains only coefficients with index less than n , so (3.4) determines c_n whenever $I_p(n) \neq 0$. The positive integer zeros of I_p are the

resonances of the recurrence: at such an index the recurrence leaves c_n undetermined if its right side vanishes, and has no solution otherwise.

If p is odd, then $c_* = 2R_p$, and there is no positive integer resonance. For integers $1 \leq n < p$, the product in (3.3) is negative with absolute value less than R_p . For $p \leq n \leq 2p - 1$ it is zero, and for $n \geq 2p$ it is positive. Hence $I_p(n) \neq 0$ for every positive integer n , and there is exactly one formal Laurent expansion at the marked pole.

Theorem 2 of Eremenko [19] states that an autonomous polynomial differential equation with finitely many possible formal pole expansions and a single monomial of highest total degree has all its meromorphic solutions in $\mathcal{W}$. In (1.1) the only monomial of degree two is $v^2/2$, and for odd p we have just seen that there is exactly one formal pole expansion. Both hypotheses hold at every odd order, for every coefficient vector a .

3.2 The energy argument for an even operator

Let $p = 2m$ and

$$P(D) = D^{2m} + \sum_{j=0}^{m-1} b_j D^{2j}. \quad (3.5)$$

In this subsection b_j denotes the coefficients of the even operator, not those of (2.1). Now $c_* = -2R_p$, and there is exactly one positive integer resonance. Indeed the product in (3.3) is at most R_p for $0 \leq n < p$, is zero for $p \leq n \leq 2p - 1$, and is strictly increasing for $n \geq 2p$; its value at $3p$ is $2R_p = -c_*$. The only positive integer zero of I_p is therefore $n = 3p$, and the possible free Laurent coefficient is

$$h = c_{3p} = [z^{2p}]v.$$

If the recurrence is compatible at $n = 3p$, the equation has a one-parameter family of formal expansions at the pole, one for each value of h , and Eremenko's theorem does not apply to (1.1) itself. The remedy is a first integral whose value determines h .

For $j \geq 1$, define the quadratic differential expression

$$\mathcal{B}_j(v) = \sum_{\ell=0}^{j-2} (-1)^\ell v^{(\ell+1)} v^{(2j-1-\ell)} + \frac{(-1)^{j-1}}{2} (v^{(j)})^2, \quad (3.6)$$

where the sum is empty when $j = 1$. Termwise differentiation gives $D\mathcal{B}_j = v'v^{(2j)}$: every even derivative, multiplied by v' , is an exact derivative. Since P contains only even derivatives, it follows that

$$\mathcal{H}_P(v) = \mathcal{B}_m(v) + \sum_{j=1}^{m-1} b_j \mathcal{B}_j(v) + \frac{b_0}{2} v^2 + \frac{1}{6} v^3 \quad (3.7)$$

satisfies

$$D\mathcal{H}_P = v'E_P(v). \quad (3.8)$$

In particular $\mathcal{H}_P$, the energy, is constant on each solution. It is the energy in the usual sense: an even operator $P(D)$ is formally self-adjoint, (1.1) is then the Euler–Lagrange equation of an autonomous Lagrangian, and $\mathcal{H}_P$ is the conserved quantity that belongs to translation in z . The purely dispersive equations are also exactly those for which (1.1) is invariant under $z \mapsto -z$. The case with no intermediate derivatives is the first integral used by Eremenko et al. [20]. The familiar Kawahara first integral is the case $m = 2$ [15, 48]; the corresponding order-six construction appears in Dang [10].

The energy identity has two consequences for the Laurent expansion. The first is that the resonance is always compatible. To see this, construct (3.1) through index $3p - 1$. The first possible residual of (1.1) is $\Omega_p z^p$. Multiplication by v' gives residue $-pc_*\Omega_p$. By (3.8) this is the residue of a derivative of a Laurent series, and is therefore zero. Hence $\Omega_p = 0$, and h may be chosen arbitrarily.

The second consequence is that the energy determines h . On the full formal solution the constant energy has the form

$$\mathcal{H}_P = \Lambda_p h + \eta_p(P), \quad \Lambda_p = -pc_* I'_p(3p) \neq 0. \quad (3.9)$$

To find the coefficient, linearize the highest-weight part of (3.8) about $c_* z^{-p}$, with perturbation z^{n-p} . If the corresponding energy term is $L(n)z^{n-3p}$, then

$$(n - 3p)L(n) = -pc_* I_p(n).$$

Taking $n \rightarrow 3p$ gives the coefficient in (3.9). Lower derivative terms have different powers of z and cannot contribute to this coefficient. Moreover,

$$I'_p(3p) = 2R_p \sum_{j=p+1}^{2p} \frac{1}{j} > 0.$$

Fixing the energy thus fixes h , and all subsequent Laurent coefficients follow from (3.4). This is the resonance-removal step in Eremenko et al. [20, Lemma 4], with an explicit coefficient for the present operator family; see also Conte et al. [7, Section 3.2] and Conte et al. [8, Section 5].

We can now apply Eremenko's theorem, not to (1.1) but to the level equation $\mathcal{H}_P(v) = H$ with P and H fixed. This equation is autonomous and polynomial, and its unique monomial of highest degree is $v^3/6$. It has exactly one formal pole expansion: differentiating it and dividing by the nonzero Laurent series v' returns $E_P(v) = 0$, so every expansion belongs to the family above, and (3.9) selects the value of h . Every meromorphic solution of (1.1) lies on one energy level and therefore belongs to $\mathcal{W}$.

3.3 The period group and the entire solutions

It remains to show that the poles of one solution form a single orbit under its periods. For either covered case, take two poles z_1, z_2 of the same solution. The functions $v(z + z_1)$ and $v(z + z_2)$ satisfy the same equation and, in the even case, have the same energy. Their Laurent series at zero coincide, so the identity theorem gives

$$v(z + z_1) = v(z + z_2).$$

Every difference of poles is a period. Conversely, every period translates a pole to a pole. The period group is discrete: a sequence of nonzero periods tending to zero would force $v' \equiv 0$. The pole set is therefore one coset of that group. The same argument proves the uniqueness statements of the theorem: two nonconstant solutions with the same operator, the same energy in the even case, and a pole at zero have the same Laurent series there.

Finally, an entire solution in $\mathcal{W}$ must be constant. In the rational case it is a polynomial, and the highest degree of its square cannot be cancelled in (1.1) unless the polynomial is constant. In the rational-exponential case it is a Laurent polynomial in $t = e^{\kappa z}$; the operator $P(\kappa t \partial_t)$ preserves each exponent, whereas squaring doubles an extreme nonzero exponent. An entire

elliptic function is constant by compactness. The constants solve $a_0v + v^2/2 = 0$, which gives $v = 0$ and $v = -2a_0$. This completes the proof of Theorem 3.1.

The same uniqueness shows that, in the two pure subfamilies of Section 2, every wave inherits the reflection symmetry of its equation.

Corollary 3.2 (Waves are symmetric about their poles). *Let v be a nonconstant meromorphic solution of (1.1) and let z_0 be one of its poles.*

- (i) *If P is even, then $v(z_0 + \zeta) = v(z_0 - \zeta)$.*
- (ii) *If p is odd and $P - P(0)$ is odd, then $v(z_0 + \zeta) + P(0) = -(v(z_0 - \zeta) + P(0))$.*

Consequently a solution that is rational in an exponential is a pulse in case (i). In case (ii) its two limits add up to $-2P(0)$, so it is a front unless $P(0) = 0$.

Proof. In case (i) the function $v(z_0 - \zeta)$ solves the same equation, has a pole at $\zeta = 0$, and has the same energy as v , because every term of (3.6) and (3.7) contains an even total number of derivatives. Fixed-energy uniqueness gives (i). In case (ii) the function $-v(z_0 - \zeta) - 2P(0)$ solves the same equation, by Section 2, and has a pole at $\zeta = 0$. For odd p the Laurent expansion at a pole is unique, which gives (ii). The statements about the limits follow by letting ζ tend to the two ends. $\square$

4 Three finite algebraic systems

Theorem 3.1 says that, in the covered cases, a nonconstant meromorphic solution is one of three kinds and has one pole per period. Each kind is a finite-dimensional family of functions: a polynomial in $1/z$, a polynomial in a fixed simply periodic function with one pole, or a constant plus a linear combination of $\wp$ and its derivatives. Substituting the general member of each family into (1.1) therefore gives finitely many polynomial equations in its coefficients. This section writes down the three systems, fixes their normalizations, and proves that they are exactly equivalent to the one-pole solutions (Theorem 4.1). They are instances of the principal-part construction of Demina and Kudryashov [16, 17]; for a fixed operator the same construction reduces to the coefficient comparison of Demina and Kudryashov [12], which we do not need.

The systems all have the same shape. Let V be a polynomial expression in some variables, and let δ be a differentiation rule for those variables, so that the polynomials in them form a differential algebra with derivation δ . Write

$$\mathcal{E}_P(V; \delta) = \delta^p V + \sum_{j=0}^{p-1} a_j \delta^j V + \frac{1}{2} V^2.$$

Each system below is the polynomial identity $\mathcal{E}_P(V; \delta) = 0$ for an appropriate V and δ . A polynomial identity means the vanishing of every coefficient.

4.1 Rational solutions

A rational solution whose only pole is at zero, of order p , and which has no nonconstant polynomial part at infinity is a polynomial of degree p in $T = 1/z$. Since $DT = -T^2$, the equation becomes

$$V_R(T) = b + \sum_{k=1}^p r_k T^k, \quad r_p = c_*, \quad \mathcal{E}_P(V_R; -T^2 \partial_T) = 0. \quad (4.1)$$

The left side of the identity is a polynomial of degree at most $2p$ in T , so this is a finite system in $b, r_1, \dots, r_{p-1}$. No rational solution has a nonconstant polynomial part at infinity: such a part would have its degree doubled by v^2 , while the linear differential expression cannot reach that degree. Consequently (4.1) is the full rational one-pole system.

4.2 Rational functions of an exponential

For $q = \kappa^2 \neq 0$, define

$$T_q(z) = \frac{\kappa}{2} \coth \frac{\kappa z}{2}, \quad DT_q = \frac{q}{4} - T_q^2. \quad (4.2)$$

The expression is independent of the choice of sign of κ . It has period $2\pi i/\kappa$ and one pole per period, at zero, with residue one; it plays the role that $1/z$ plays in the rational case. The finite system is

$$V_S(T) = \sum_{k=0}^p u_k T^k, \quad u_p = c_*, \quad q \neq 0, \quad \mathcal{E}_P \left(V_S; \left(\frac{q}{4} - T^2 \right) \partial_T \right) = 0. \quad (4.3)$$

To see that every one-pole solution rational in an exponential has this form, write $v = R(t)$, $t = e^{\kappa z}$. The rational function R has finite limits at both $t = 0$ and $t = \infty$. If it had a pole at either end, an extreme Laurent monomial would be doubled by R^2 and could not cancel against $P(\kappa t \partial_t)R$. After a translation, its only finite pole is at $t = 1$, of order p . Such a function is a degree- p polynomial in T_q . The logistic coordinate

$$Q = \frac{1}{1 + e^{-z}}, \quad DQ = Q(1 - Q),$$

is an equivalent coordinate after fixing the rate and translating the pole; $Q(z + i\pi) = T_1(z) + 1/2$. Polynomial logistic representations and their use in constructing travelling waves are developed by Kudryashov [31, 35, 36].

The rational system is the $q = 0$ specialization of (4.3), with $T_0 = 1/z$. We keep $q = 0$ and $q \neq 0$ distinct so that this degeneration is never counted twice.

4.3 Elliptic solutions

Use the Weierstrass algebra

$$Y^2 = 4X^3 - g_2X - g_3, \quad \mathcal{D}X = Y, \quad \mathcal{D}Y = 6X^2 - \frac{g_2}{2}, \quad \Delta = g_2^3 - 27g_3^2 \neq 0. \quad (4.4)$$

Its analytic realization is $X = \wp(z; g_2, g_3)$, $Y = \wp'(z; g_2, g_3)$; see NIST [45, Chapter 23]. Set

$$V_E = b + \sum_{j=0}^{p-2} t_j \mathcal{D}^j X, \quad t_{p-2} = -s_p, \quad s_p = \frac{2(2p-1)!}{((p-1)!)^2}, \quad \mathcal{E}_P(V_E; \mathcal{D}) = 0. \quad (4.5)$$

Using (4.4), every polynomial in X and Y reduces to the normal form $F(X) + YG(X)$. The system consists of the coefficients of F and G set equal to zero; all of them are polynomials with rational coefficients in the displayed unknowns.

The representation is the classical one. An elliptic function with one pole has zero residue there. The functions $1, \wp, \wp', \dots, \wp^{(p-2)}$ have distinct principal parts and span the space of elliptic functions whose only pole is at zero, of order at most p . Subtracting the indicated principal part

from a one-pole solution leaves an entire elliptic function, hence a constant. The leading coefficient in (4.5) follows from

$$D^{p-2}\wp = (-1)^{p-2}(p-1)!z^{-p} + O(1).$$

The nonsingularity condition $\Delta \neq 0$ is retained throughout; the singular cubics correspond to the rational and simply periodic systems, which we have already written down.

Theorem 4.1 (Finite matching and exhaustion). *For every $p \geq 2$, the systems (4.1), (4.3), and (4.5) are necessary and sufficient for the respective one-pole solutions of (1.1), with the pole at zero and with the stated normalizations. For the operator classes of Theorem 3.1, these systems, together with the constant solutions, exhaust all globally meromorphic solutions.*

The systems describe exceptional parameter values and continuous families as well as isolated solutions. Their sufficiency follows from substitution, without any genericity assumption.

Proof. Necessity is the principal-part argument given for each kind above. Conversely, a solution of a coefficient system gives a globally meromorphic function by its displayed realization, and the chain rule gives $E_P(v) = 0$. The leading coefficient $c_* \neq 0$ ensures a nonconstant function. In the elliptic case every nonsingular pair (g_2, g_3) has a Weierstrass realization, and the normal form $F(X) + YG(X)$ is unique. The exhaustion statement is Theorem 3.1. $\square$

5 The count of rational-exponential pairs

In this section and the next, P ranges over *all* monic polynomials of degree p , mixed operators included, and we count the pairs (P, v) of Section 2. Nothing in the count requires the hypotheses of Theorem 3.1. Those hypotheses were needed to show that the one-pole solutions are all the meromorphic solutions; the matching equations and their solution counts make sense without them. The pairs that belong to the purely dispersive and purely dissipative subfamilies are counted in Section 5.4.

Recall the normalization: the rate is one, the pole is at $i\pi$ modulo $2\pi i\mathbb{Z}$, and $v \rightarrow 0$ as $e^z \rightarrow 0$. A wave and its mirror image are counted separately when they differ. The functions $Q, DQ, \dots, D^{p-1}Q$ span the degree- p polynomials in Q with zero constant term, since

$$D^j Q = (-1)^j j! Q^{j+1} + \text{lower powers of } Q.$$

The leading pole coefficient (3.2) therefore gives the unique form

$$v_A = s_p A(D)Q, \quad A(\rho) = \rho^d + \alpha_1 \rho^{d-1} + \dots + \alpha_d, \quad d = p - 1. \quad (5.1)$$

The scale s_p was defined in (4.5). This representation is the standard singular-part operator form: $Q = 1/2 + D \log \cosh(z/2)$, so (5.1) is a constant plus a linear differential operator applied to $\log \cosh(z/2)$. Such forms, including the KS case, are discussed by Conte and Musette [4]. We use the coefficients $\alpha_1, \dots, \alpha_d$ of that operator as the coordinates for the enumeration.

The argument runs as follows. The profile v_A solves (1.1) for some monic P exactly when A divides a polynomial built from A by a discrete convolution (Proposition 5.1). The remainder of that division gives d polynomial equations in the d coordinates. We count their solutions with multiplicity by Bézout's theorem, using a weighted degree adapted to the equations (Theorem 5.3). When $2p - 1$ is prime, reduction modulo $2p - 1$ identifies the solutions one by one (Theorem 5.4). The reflection $z \mapsto -z$ then singles out the pairs of the two pure subfamilies (Theorem 5.8).

5.1 Discrete convolution and the matching equations

For polynomials F, G , define $F \star G$ by polynomial continuation of

$$(F \star G)(n) = \sum_{j=1}^{n-1} F(j)G(n-j), \quad n \geq 1.$$

The continuation exists by the power-sum formulas: for fixed F, G the right side is a polynomial in n . Write

$$S_A = A \star A.$$

For monic A of degree d ,

$$\deg S_A = 2d + 1, \quad [\rho^{2d+1}]S_A = \beta_d := \frac{(d!)^2}{(2d+1)!}. \quad (5.2)$$

The leading coefficient is the beta integral $\int_0^1 t^d(1-t)^d dt$.

Proposition 5.1 (Polynomial divisibility). *The profile (5.1) solves (1.1) for some monic P if and only if*

$$A \mid S_A. \quad (5.3)$$

The operator is unique and is

$$P_A = \frac{s_p S_A}{2A}. \quad (5.4)$$

Proof. In the half-plane $\operatorname{Re} z < 0$, the logistic function is a geometric series in e^z , and applying $A(D)$ multiplies the n th term by $A(n)$:

$$A(D)Q = \sum_{n \geq 1} (-1)^{n-1} A(n) e^{nz}.$$

Squaring gives the rational-function identity

$$(A(D)Q)^2 = -S_A(D)Q. \quad (5.5)$$

The map $B \mapsto B(D)Q$ is injective: its vanishing would give $B(n) = 0$ for every positive integer n . Consequently (1.1) is equivalent to $2P_A A = s_p S_A$, which has a polynomial solution P_A exactly when A divides S_A . Equation (5.2) and $s_p = 2/\beta_d$ show that P_A is monic of degree p . $\square$

The operator determines the profile. This holds for every p , without the hypotheses of Theorem 3.1, because it only uses the leading coefficient of a convolution.

Proposition 5.2 (One profile per operator). *Let A, B be monic polynomials of degree $p-1$ satisfying $A \mid S_A$ and $B \mid S_B$. If $P_A = P_B$, then $A = B$. Consequently a monic operator of degree p admits at most one normalized rational-exponential one-pole profile. Any count of distinct pairs with this normalization is therefore also a count of distinct operators.*

Proof. Suppose $P_A = P_B = P$, and put $h = A - B$. If $h \neq 0$, its degree e is less than $d = p-1$; write $f \neq 0$ for its leading coefficient. By (5.4) and the bilinearity and symmetry of the convolution,

$$S_A - S_B = h \star (A + B) = \frac{2}{s_p} Ph.$$

The power-sum calculation in (5.2) gives the leading coefficient $2f \int_0^1 t^e(1-t)^d dt$ on the left. Since P is monic, the right side has leading coefficient $(2/s_p)f = \beta_d f$. Cancelling f would therefore give

$$2 \int_0^1 t^e(1-t)^d dt = \int_0^1 t^d(1-t)^d dt.$$

This is impossible: $e < d$ implies $t^e > t^d$ for $0 < t < 1$, so the left side exceeds twice the right side. Thus $h = 0$. $\square$

Divide S_A by the monic polynomial A :

$$S_A = AC_A + R_A, \quad R_A = \sum_{r=0}^{d-1} R_r(\alpha_1, \dots, \alpha_d) \rho^r. \quad (5.6)$$

The matching equations are $R_0 = \dots = R_{d-1} = 0$: d polynomial equations in the d unknowns $\alpha_1, \dots, \alpha_d$. Their solutions are in bijection with the normalized pairs (P_A, v_A) . We write $\mathcal{X}_p$ for the solution set together with the equations that define it; in algebraic language, $\mathcal{X}_p$ is the affine scheme defined by the R_r .

Three notions from algebra are used to count. A solution is *simple* if the Jacobian matrix of $(R_0, \dots, R_{d-1})$ with respect to $(\alpha_1, \dots, \alpha_d)$ is invertible there. Every isolated solution has a *multiplicity*. For a square system such as this one it may be pictured as the number of simple solutions into which the point splits under a generic small perturbation of the equations; formally it is the dimension over $\mathbb{C}$ of the local quotient algebra of the equations at that point. It equals one exactly for simple solutions. When $\mathcal{X}_p$ is finite, its *length* is the sum of the multiplicities of its points, and $\mathcal{X}_p$ is *reduced* when every point is simple, so that the length is the number of points. In these terms, a count with multiplicity is a statement about the length of $\mathcal{X}_p$, and the statement that all pairs are distinct means that $\mathcal{X}_p$ is reduced.

5.2 Weighted degree and the role of continuous convolution

Bézout's theorem bounds the sum of the multiplicities of the isolated solutions of d polynomial equations in d unknowns by the product of their degrees. Equality holds when the system has no solutions at infinity, that is, when the highest-degree parts of the equations have no common zero other than the origin. The ordinary degrees of the matching equations give a poor bound. The right measure of their size is a weighted degree that reflects how the α_j scale. Give α_j weight j , and ρ weight one. The remainder equations have highest weighted degrees

$$2d + 1 - r \quad (0 \leq r < d), \quad \text{namely } d + 2, \dots, 2d + 1. \quad (5.7)$$

Their highest weighted parts are the coefficients of $\text{rem}_A H_A$, where

$$H_A(\rho) = \int_0^\rho A(t)A(\rho - t) dt. \quad (5.8)$$

Indeed, if $A(\rho) = \sum a_i \rho^i$, then

$$H_A(\rho) = \sum_{i,j=0}^d \frac{i!j!}{(i+j+1)!} a_i a_j \rho^{i+j+1}.$$

These are precisely the highest weighted terms of the power sums defining S_A , and monic division preserves the highest weighted part. Indeed, A is weighted-homogeneous of weight d , and each

step of long division subtracts a multiple of A of the same weight as the term being removed. In words, the discrete convolution S_A is a lower-order perturbation of the continuous convolution H_A , and the behaviour of the matching equations at infinity is governed by the continuous problem $A \mid H_A$.

For this degree, denote by (R_d) the condition

$$A \text{ monic, } \deg A = d, \quad A \mid H_A \implies A(\rho) = \rho^d. \quad (5.9)$$

It says that the continuous problem has only the trivial solution. By the preceding paragraph, it is exactly the condition that the matching equations have no solution at weighted infinity.

Theorem 5.3 (Count with algebraic multiplicities). *For every $p \geq 2$, the sum of multiplicities of the isolated points of $\mathcal{X}_p$ is at most*

$$N_p = \binom{2p-1}{p-1}.$$

If (R_{p-1}) holds, then $\mathcal{X}_p$ is finite and

$$\text{length } \mathcal{X}_p = N_p. \quad (5.10)$$

This equality does not require simplicity.

Proof. Weighted degree is turned into ordinary degree by the finite substitution

$$\alpha_j = y_j^j, \quad 1 \leq j \leq d. \quad (5.11)$$

A monomial of weight w in the α_j becomes a monomial of ordinary degree w in the y_j , so the pulled-back equations have ordinary degrees at most $d+2, \dots, 2d+1$, and their highest-degree parts are the pull-backs of the highest weighted parts. Bézout's inequality for isolated solutions bounds the sum of their multiplicities by $(d+2) \cdots (2d+1)$; see Mondal [42, Chapter VIII]. If (R_d) holds, the leading homogeneous forms have no nonzero common zero. The homogenized system has no point at infinity, and ordinary Bézout gives equality [9].

It remains to return from the y_j to the α_j . A solution in the α_j with all $\alpha_j \neq 0$ has $1 \cdot 2 \cdots d = d!$ preimages in the y_j , one for each choice of the roots $y_j = \alpha_j^{1/j}$, each with the original multiplicity. In general, for each solution the sum of the multiplicities of its preimages is $d!$ times its original multiplicity, including at branch points where some α_j vanish. Thus the substitution multiplies the total length by $d!$. Algebraically, $\mathbb{C}[y_1, \dots, y_d]$ is a free module over $\mathbb{C}[\alpha_1, \dots, \alpha_d]$, with basis

$$y_1^{e_1} \cdots y_d^{e_d}, \quad 0 \leq e_j < j.$$

This remains true after imposing the equations and restricting to an isolated component. Dividing the Bézout count by $d!$ gives

$$\frac{(d+2) \cdots (2d+1)}{d!} = \binom{2d+1}{d}.$$

The no-infinity condition also gives finiteness of the whole affine scheme, completing the proof. $\square$

For $p = 3$, the variable weights are 1, 2 and the equation weights are 4, 5, so the calculation is $(4 \cdot 5)/(1 \cdot 2) = 10$.

5.3 The prime-index theorem

Theorem 5.4 (Prime-index enumeration). *If $\ell = 2p - 1$ is prime, then*

$$\#\mathcal{X}_p(\mathbb{C}) = \text{length } \mathcal{X}_p = \binom{\ell}{p-1},$$

and every point is simple. After a choice of an ℓ -adic embedding of the algebraic coefficients, reduction labels the points by the $(p-1)$ -element subsets $S \subset \mathbb{F}_\ell$:

$$\overline{A}(\rho) = \prod_{s \in S} (\rho - s). \quad (5.12)$$

Proof. Put $d = p - 1$, so $\ell = 2d + 1$, and work over $\mathbb{Z}_{(\ell)}$, the rationals integral at ℓ . The proof has four steps: a congruence for S_A modulo ℓ ; the solution of the reduced equations over $\mathbb{F}_\ell$; the exclusion of solutions at infinity, which proves (R_d) ; and the lifting of the reduced solutions back to characteristic zero.

Step 1: the congruence. Induction using $(D - n)Q^n = -nQ^{n+1}$ gives

$$Q^n = \frac{(-1)^{n-1}}{(n-1)!} \prod_{j=1}^{n-1} (D - j)Q.$$

Write $(A(D)Q)^2 = \sum_{n=2}^{2p} b_n Q^n$. Each b_n is an integer polynomial in the coefficients of A , and $b_{2p} = (d!)^2$. By (5.5),

$$S_A(\rho) = \sum_{n=2}^{2p} \frac{(-1)^n b_n}{(n-1)!} \prod_{j=1}^{n-1} (\rho - j).$$

Only the last denominator, $(2p-1)! = \ell!$, contains ℓ . Consequently ℓS_A is integral and

$$\ell S_A(\rho) \equiv \frac{(d!)^2}{(\ell-1)!} \prod_{j=1}^{\ell} (\rho - j) \equiv (-1)^d (\rho^\ell - \rho) \pmod{\ell}. \quad (5.13)$$

The scalar follows by pairing j with $\ell - j$ in $(\ell-1)!$, and the product is $\rho^\ell - \rho$ because it runs over all residues modulo ℓ .

Step 2: the reduced equations. The scaled remainder equations $\ell R_r = 0$ therefore reduce exactly to $\overline{A} \mid \rho^\ell - \rho$. There are $\binom{\ell}{d}$ monic divisors of degree d , all defined over $\mathbb{F}_\ell$, and these exhaust the solutions over its algebraic closure. Each of them is a simple solution of the reduced equations. Write $\rho^\ell - \rho = \overline{A}C$. For a variation h with $\deg h < d$, the derivative of the remainder map is, up to the nonzero scalar $(-1)^d$,

$$h \longmapsto -\text{rem}_{\overline{A}}(Ch). \quad (5.14)$$

Since $\rho^\ell - \rho$ is squarefree, $\overline{A}$ and C are coprime, so multiplication by C is invertible modulo $\overline{A}$, and the map (5.14) is invertible.

Step 3: no solutions at infinity. We must also prove that characteristic-zero solutions cannot escape to infinity, that is, that (R_d) holds. Taking the highest weighted part of (5.13) gives

$$\ell H_A(\rho) \equiv (-1)^d \rho^\ell \pmod{\ell}. \quad (5.15)$$

Thus the reduced leading remainder equations have only $\overline{A} = \rho^d$. After (5.11), they are homogeneous equations with no nonzero common zero over $\overline{\mathbb{F}}_\ell$. The same is true in characteristic zero.

If the equations had a complex projective zero, they would also have one with algebraic coordinates, since their coefficients are rational. Normalization at an ℓ -adic place would then produce a nonzero reduced point. The valuation argument is stated precisely in Appendix A. We have proved (R_d) , so Theorem 5.3 gives total length $\binom{\ell}{d}$.

Step 4: lifting. The multivariable simple-root Hensel lemma lifts each of the $\binom{\ell}{d}$ reduced points to a unique point over $\mathbb{Z}_\ell$; see Conrad [3]. The lifts are distinct and simple. Their number already equals the characteristic-zero total length, so they exhaust the scheme. The equations define a finite algebra over $\mathbb{Q}$, hence every coordinate of a solution is algebraic over $\mathbb{Q}$, and the number of geometric points and their local lengths are the same over $\overline{\mathbb{Q}_\ell}$ and over $\mathbb{C}$. This proves the assertion and the labelling. $\square$

Remark 5.5 (Formal verification). The count and simplicity assertions of Theorem 5.4, and (R_{p-1}) for $2p-1$ prime, have been verified in Lean 4 against the Mathlib library; Appendix A describes the files and the scope of the verification. The formal proof follows a different route, which does not use Theorem 5.3. A valuation argument shows that every complex matching point is integral at ℓ and that reduction modulo ℓ is injective on these points; this gives finiteness, the upper bound $\binom{\ell}{d}$ and simplicity. The lower bound comes from Hensel lifting of the reduced points. The labelling (5.12) is the injective reduction map of that proof, and the repository documents the route in detail.

The labels (5.12) describe reductions of algebraic coefficients; they are not assertions that the complex roots of A , or the characteristic roots of P_A , are integers. For example, $A = \rho^2 + 1/11$ at $p = 3$ reduces to $(\rho - 2)(\rho - 3)$ modulo five.

The endpoints of v_A are its limits as $z \rightarrow \mp\infty$, namely 0 and $s_p A(0)$. Pairs with $A(0) = 0$ have equal endpoints and are pulses; the others are fronts. The pulses are needed separately, because the elliptic count of Section 6 degenerates onto them.

Proposition 5.6 (Equal endpoints). *Assume (R_{p-1}) , and let $\mathcal{Z}_p = \mathcal{X}_p \cap \{A(0) = 0\}$, with its induced scheme structure. Then*

$$\text{length } \mathcal{Z}_p = S_p := \binom{2p-2}{p-2}. \quad (5.16)$$

If $\mathcal{X}_p$ is reduced, these are S_p distinct pairs and $P_A(0) \neq 0$ at each of them. In particular, under the prime-index hypothesis, exactly

$$\binom{2p-2}{p-2}$$

normalized pairs have equal zero endpoints. The remaining $\binom{2p-2}{p-1}$ have unequal endpoints. These are counts over $\mathbb{C}$.

Proof. Polynomial summation gives $S_A(0) = -A(0)^2$, so the constant remainder satisfies the polynomial identity

$$R_0 = -A(0)(A(0) + C_A(0)). \quad (5.17)$$

Set $\alpha_d = A(0) = 0$. The equation $R_0 = 0$ disappears, leaving $d-1$ equations of weights $d+2, \dots, 2d$ in variables of weights $1, \dots, d-1$. Their leading forms have no nonzero common zero by (R_d) . The weighted calculation in Theorem 5.3 gives

$$\text{length } \mathcal{Z}_p = \frac{(d+2) \cdots (2d)}{(d-1)!} = \binom{2d}{d-1}.$$

For $d = 1$ the system is empty and its length is one.

If $\mathcal{X}_p$ is finite and reduced, its coordinate algebra is a product of copies of $\mathbb{C}$; the same is true of any quotient, in particular that of $\mathcal{Z}_p$, so $\mathcal{Z}_p$ is reduced as well. At a point with $A(0) = C_A(0) = 0$, the differential of (5.17) vanishes. Such a point cannot be simple in the square matching system. Therefore $C_A(0) \neq 0$, and $P_A(0) = s_p C_A(0)/2 \neq 0$, at every point of $\mathcal{Z}_p$ when $\mathcal{X}_p$ is reduced. The prime-index assertions now follow from Theorem 5.4 and $N_p - S_p = \binom{2p-2}{p-1}$. $\square$

Under the labelling (5.12) of Theorem 5.4, a pair is a pulse exactly when its label set contains 0. Indeed $A(0) = 0$ forces 0 to be a root of the reduced polynomial $\bar{A}$, and the two counts agree: the $(p-1)$ -element subsets of $\mathbb{F}_\ell$ containing 0 number $\binom{2p-2}{p-2} = S_p$, and those avoiding 0 number $\binom{2p-2}{p-1} = N_p - S_p$.

5.4 Reflection: the two pure subfamilies

The numbers N_p and M_p refer to the full monic family. We now count the pairs whose equation is purely dispersive or purely dissipative. These are the fixed points of the reflection $z \mapsto -z$ acting on the pairs, and their number is again a binomial coefficient.

For a monic polynomial A of degree $d = p - 1$ put

$$(\iota A)(\rho) = (-1)^d A(-\rho), \quad \text{that is,} \quad \alpha_j \mapsto (-1)^j \alpha_j. \quad (5.18)$$

Proposition 5.7 shows that ι is the reflection $z \mapsto -z$ written in the coordinates α_j . It reads the wave v_A in the opposite direction and then restores the normalization: it moves the zero limit back to the end $e^z \rightarrow 0$, and for odd p it changes the sign of the profile.

Proposition 5.7 (Reflection). *For every monic A of degree d ,*

$$S_{\iota A}(\rho) = -S_A(-\rho) - 2A(0)A(-\rho), \quad R_{\iota A}(\rho) = -R_A(-\rho). \quad (5.19)$$

Consequently ι is an involution of the scheme $\mathcal{X}_p$. At a point of $\mathcal{X}_p$,

$$v_{\iota A}(z) = (-1)^p (v_A(-z) - s_p A(0)), \quad P_{\iota A}(\rho) = (-1)^p (P_A(-\rho) + s_p A(0)). \quad (5.20)$$

The pairs A and ιA are the mirror images that N_p counts separately. For $p = 3$ the involution is $(a, b) \mapsto (-a, b)$ in Table 3.

Proof. Since $Q(-z) = 1 - Q(z)$, and D changes sign under $z \mapsto -z$,

$$(A(D)Q)(-z) = A(0) - A(-D)Q = A(0) - (-1)^d (\iota A)(D)Q.$$

This is the first formula in (5.20). Evaluate (5.5) at $-z$, and use (5.5) for ιA on the left:

$$A(0)^2 - 2A(0)A(-D)Q - S_{\iota A}(D)Q = -S_A(0) + S_A(-D)Q.$$

Every $B(D)Q$ is a polynomial in Q without constant term, so letting $z \rightarrow -\infty$ gives $S_A(0) = -A(0)^2$. The remaining terms, and the injectivity of $B \mapsto B(D)Q$, give the first identity in (5.19). Now substitute $S_A = AC_A + R_A$ at $-\rho$:

$$S_{\iota A}(\rho) = (\iota A)(\rho) (-1)^p (C_A(-\rho) + 2A(0)) - R_A(-\rho).$$

Division by the monic polynomial ιA is unique. Hence

$$R_{\iota A}(\rho) = -R_A(-\rho), \quad C_{\iota A}(\rho) = (-1)^p (C_A(-\rho) + 2A(0)).$$

The first relation reads $R_r(\iota \alpha) = (-1)^{r+1} R_r(\alpha)$, so the linear map (5.18) preserves the ideal of the matching equations. The second relation and (5.4) give the formula for $P_{\iota A}$. $\square$

Let

$$\mathcal{X}_p^{\text{sym}} = \mathcal{X}_p \cap \{\alpha_j = 0 \text{ for odd } j\}$$

be the fixed locus of ι , with its induced scheme structure, and put

$$F_p = \binom{p-1}{\lfloor (p-1)/2 \rfloor}. \quad (5.21)$$

Its first values are 1, 2, 3, 6, 10, 20 for $p = 2, \dots, 7$; Table 1 lists the equations to which they belong.

Theorem 5.8 (Count in the two pure subfamilies). (i) *A point A of $\mathcal{X}_p$ lies in $\mathcal{X}_p^{\text{sym}}$ if and only if P_A is reflection-symmetric. Thus $\mathcal{X}_p^{\text{sym}}$ consists of the rational-exponential pairs of the purely dispersive equations when p is even, and of the purely dissipative equations when p is odd.*

(ii) *If (R_{p-1}) holds, then $\text{length } \mathcal{X}_p^{\text{sym}} = F_p$.*

(iii) *If, in addition, $\mathcal{X}_p$ is reduced, then $\mathcal{X}_p^{\text{sym}}$ consists of F_p distinct pairs, each a simple solution of the matching equations of its subfamily. If $2p-1$ is prime, they are the points whose label in (5.12) satisfies $S = -S$.*

(iv) *For even p , every pair of $\mathcal{X}_p^{\text{sym}}$ is a pulse. For odd p , every pair of $\mathcal{X}_p^{\text{sym}}$ is a front if $\mathcal{X}_p$ is reduced, and every pair with real coefficients is a front without any hypothesis.*

In plain terms, the pairs that coincide with their mirror image are exactly the pairs of the purely dispersive equations, for even p , or of the purely dissipative equations, for odd p . Counted with multiplicity there are F_p of them whenever rigidity holds, in particular for $p \leq 73$. They are F_p distinct pairs whenever all N_p pairs for that p are distinct, in particular when $2p-1$ is prime. The rational-exponential waves of a purely dispersive equation are pulses, for every even p . Those of a purely dissipative equation are fronts; this is proved whenever all N_p pairs are distinct, and for waves with real coefficients at every odd p . Elliptic waves occur in both subfamilies; the distinction between pulses and fronts does not apply to them.

Proof. (i). If $\iota A = A$, then (5.20) gives $P_A(\rho) = (-1)^p(P_A(-\rho) + s_p A(0))$. For even p the polynomial A is odd, so $A(0) = 0$ and P_A is even. For odd p the sum $P_A(\rho) + P_A(-\rho)$ is the constant $-s_p A(0)$. In both cases P_A is reflection-symmetric, and

$$s_p A(0) = -2P_A(0) \quad (p \text{ odd}). \quad (5.22)$$

Conversely, let $P = P_A$ be reflection-symmetric, and apply Corollary 3.2 at the pole $i\pi$, using the period $2\pi i$. If p is even, then $v_A(-z) = v_A(z)$. Letting $z \rightarrow -\infty$ gives $A(0) = 0$, and (5.20) becomes $v_{\iota A} = v_A$. If p is odd, then $-v_A(-z) = v_A(z) + 2P(0)$, and (5.20) becomes $v_{\iota A}(z) = v_A(z) + 2P(0) + s_p A(0)$. Letting $z \rightarrow -\infty$ shows that the constant vanishes. In both cases $v_{\iota A} = v_A$, hence $\iota A = A$ by the uniqueness of the form (5.1).

(ii). On the subspace $\{\alpha_j = 0, j \text{ odd}\}$ we have $\iota A = A$, and (5.19) shows that R_A is an odd polynomial. The equations $R_r = 0$ with r even therefore hold identically there. With $m = \lfloor d/2 \rfloor$, there remain the m equations $R_1, R_3, \dots, R_{2m-1}$ in the m unknowns $\alpha_2, \alpha_4, \dots, \alpha_{2m}$. By (5.7) their weights are

$$\begin{array}{l|l} \text{variables} & 2, 4, \dots, 2m \\ \text{equations} & 2d, 2d-2, \dots, 2(d-m+1). \end{array}$$

Their highest weighted parts are the restrictions of those of the full system. The highest weighted parts of the omitted equations vanish identically on the subspace, because H_A is an odd polynomial when $\iota A = A$. A common zero of the restricted leading forms is therefore a polynomial A with $A \mid H_A$, and (R_d) gives $A = \rho^d$. The finite-cover argument of Theorem 5.3, with $\alpha_{2i} = y_i^{2i}$, gives

$$\text{length } \mathcal{X}_p^{\text{sym}} = \prod_{i=1}^m \frac{2(d-i+1)}{2i} = \binom{d}{m}.$$

(iii). A closed subscheme of a finite reduced scheme is reduced, as in the proof of Proposition 5.6, and a reduced point of a square system is a simple solution of it. The map (5.18) negates the roots of $\bar{A}$, so it acts on the labels by $S \mapsto -S$, and the labelling is a bijection.

(iv). For even p the polynomial A is odd, so $A(0) = 0$. Let p be odd, and suppose that a point of $\mathcal{X}_p^{\text{sym}}$ is a pulse, $A(0) = 0$. Then $P_A(0) = 0$ by (5.22). If $\mathcal{X}_p$ is reduced, this contradicts Proposition 5.6. If A has real coefficients, then v_A is real on the real axis, and it tends to zero with all its derivatives at both ends. Integrating (1.1) along the real axis, and using $P_A(0) = 0$, gives $\int v_A^2 dz = 0$. Hence $v_A = 0$, which is impossible. $\square$

The label count agrees with (5.21). The nonzero residues modulo $\ell = 2d + 1$ form d pairs $\{s, -s\}$. A set $S = -S$ of d elements consists of $d/2$ pairs if d is even, and of 0 and $(d-1)/2$ pairs if d is odd, which gives $\binom{d}{\lfloor d/2 \rfloor}$ sets in both cases. It contains 0 exactly when d is odd, in agreement with part (iv).

6 The count of elliptic pairs

We now count the elliptic pairs of Section 2. Fix a lattice $\Lambda \subset \mathbb{C}$, including its scale, and put $g_i = g_i(\Lambda)$. We place the pole at zero, allow P to vary over all monic polynomials of degree p , and retain the constant term of the profile. Neither reflection nor a lattice automorphism is factored out. By Section 4, every profile has the form

$$V = -s_p \mathcal{D}^{p-2} X + \sum_{j=0}^{p-3} u_j \mathcal{D}^j X + b. \quad (6.1)$$

Its only pole on $\mathbb{C}/\Lambda$ is zero, of order p . In particular Λ is its full period lattice. Let $\mathcal{E}_p(g_2, g_3)$ denote the matching equations obtained by substituting (6.1) in (1.1), in the unknowns $a_0, \dots, a_{p-1}$, $u_0, \dots, u_{p-3}$ and b , together with their solution set. The operator coefficients may equivalently be eliminated first, as in the proof below. Length, simplicity and reducedness are understood as in Section 5.

Theorem 6.1 (Fixed-lattice elliptic enumeration). *Assume (R_{p-1}) . For every fixed nonsingular lattice,*

$$\text{length } \mathcal{E}_p(g_2, g_3) = M_p := 2 \binom{2p-2}{p-2}. \quad (6.2)$$

If, in addition, $\mathcal{X}_p$ is reduced, then $\mathcal{E}_p(g_2, g_3)$ is reduced for (g_2, g_3) in a nonempty Zariski-open subset of $\{g_2^3 - 27g_3^2 \neq 0\}$. In particular, if $2p-1$ is prime, a generic fixed lattice admits exactly M_p normalized elliptic operator-profile pairs, all simple.

Thus the elliptic assertion on an arbitrary lattice is a count with multiplicities. Exceptional lattices, on which distinct pairs merge, do occur and cannot be discarded when counting distinct solutions; Section 9.2 determines them at order three.

The proof has five steps. We eliminate the operator, which leaves $p - 1$ equations in the $p - 1$ profile unknowns. We assign weights to the unknowns, under which the highest weighted parts of the equations are obtained by setting $g_2 = g_3 = 0$. We show that at this cusp the equations have only the trivial common zero; this is where (R_{p-1}) enters, and it means that there are no solutions at weighted infinity for any g_2, g_3 . Weighted Bézout then gives the length M_p on every lattice. Finally, at a nodal cubic the elliptic problem becomes the equal-endpoint case of the rational-exponential problem, whose solutions are simple by Proposition 5.6 when $\mathcal{X}_p$ is reduced, and simplicity spreads from the nodal parameter value to nearby nonsingular parameters.

Proof. Step 1: eliminating the operator. The coefficient at pole order $2p$ vanishes by the choice of s_p . The coefficients at orders $2p - 1, \dots, p$ determine $a_{p-1}, \dots, a_0$ successively. At each step the coefficient of the new a_j is the nonzero rational leading coefficient of $D^j V$. This is a polynomial elimination, valid over any $\mathbb{Q}$ -algebra. The residual has pole order at most $p - 1$ and zero residue. It lies in the span of

$$\{1\} \cup \{\mathcal{D}^j X : 0 \leq j \leq p - 3\}.$$

Hence $p - 1$ equations remain in the $p - 1$ profile coefficients u_j, b .

Step 2: weights. Give D, X, g_2, g_3 weights $1, 2, 4, 6$, respectively, and give the profile weight p . Then u_j has weight $p - j - 2$, and b has weight p . After the triangular elimination, the residual principal-part coefficients and the residual Laurent constant have weights

$$\begin{array}{c|c} \text{variables} & 1, 2, \dots, p - 2, p \\ \text{equations} & p + 1, p + 2, \dots, 2p - 2, 2p. \end{array} \quad (6.3)$$

The Laurent expansions and the eliminations are polynomial in g_2, g_3 , so these statements extend to the singular cubics. For fixed g_2, g_3 , the highest weighted parts of the equations in the profile variables are obtained by setting $g_2 = g_3 = 0$, since g_2 and g_3 carry positive weight.

Step 3: the cusp. At the cusp $g_2 = g_3 = 0$, $X = z^{-2}$ and $Y = -2z^{-3}$, so the profile is a rational function. Write $V = b + f(z)$, where f has its sole pole at zero and vanishes at infinity. The constant and decaying parts of the equation give

$$a_0 b + \frac{1}{2} b^2 = 0, \quad (P + b)(D)f + \frac{1}{2} f^2 = 0.$$

The leading term of f at infinity forces $a_0 + b = 0$. It follows that $b = a_0 = 0$. Under the polynomial Laplace correspondence

$$\sum_k A_k \rho^k \mapsto \sum_k k! A_k z^{-k-1},$$

differentiation corresponds to multiplication by $-\rho$, and multiplication of the rational functions corresponds to continuous convolution. After a nonzero scalar normalization, the equation for f therefore gives $A \mid H_A$ for a monic A of degree $p - 1$. By (R_{p-1}) , $A = \rho^{p-1}$. Thus $f = c_* z^{-p}$, and all the variables in (6.1) vanish. The leading equations have no nonzero common zero.

Step 4: the length. The same finite-cover Bézout argument as in Theorem 5.3, now with the weights (6.3), gives

$$\text{length } \mathcal{E}_p(g_2, g_3) = \frac{(p+1)(p+2) \cdots (2p-2)(2p)}{(p-2)! p} = 2 \binom{2p-2}{p-2}.$$

For $p = 3$, this is $(4 \cdot 6)/(1 \cdot 3) = 8$. Empty products are one, so the calculation includes $p = 2$. It proves this length at every (g_2, g_3) , including the singular parameters used below; at the cusp itself the whole length sits at the single solution where every unknown vanishes. It also makes the whole coefficient family finite over $\mathbb{C}[g_2, g_3]$: sufficiently large weighted monomials reduce to bounded weight using the leading equations, which are independent of g_2, g_3 . This finiteness is what allows simplicity to be transferred from the nodal parameter value to nearby nonsingular parameters in the last step.

Step 5: the node. To prove generic simplicity, consider the nodal parameters

$$g_2 = \frac{1}{12}, \quad g_3 = -\frac{1}{216}. \quad (6.4)$$

In the logistic coordinate they are realized by

$$X = Q^2 - Q + \frac{1}{12}, \quad Y = (2Q - 1)Q(1 - Q), \quad DQ = Q(1 - Q).$$

The nodal coordinate ring is

$$\mathbb{C}[X, Y] = \{f \in \mathbb{C}[Q] : f(0) = f(1)\}.$$

Indeed the right side is $\mathbb{C} + Q(1 - Q)\mathbb{C}[Q]$; it is generated by X, Y , since, with $S = Q(1 - Q)$, $S = 1/12 - X$, $SQ = (Y + S)/2$, and $Q^2 = Q - S$. The pole filtration is the ordinary polynomial degree in Q . Consequently the nodal specialization of (6.1) has the unique form

$$V = \beta + v_A, \quad A \text{ monic of degree } p - 1, \quad A(0) = 0,$$

where β is the common endpoint. The coefficient change is invertible and affine over $\mathbb{Q}$.

The matching equations become, as equations of schemes,

$$A \in \mathcal{Z}_p, \quad P = P_A - \beta, \quad \beta(\beta - 2P_A(0)) = 0. \quad (6.5)$$

To see this directly, the constant equation is $P(0)\beta + \beta^2/2 = 0$. The remaining polynomial identity is $(P + \beta)(D)v_A + v_A^2/2 = 0$, equivalent by Proposition 5.1 to $A \mid S_A$ and $P + \beta = P_A$. The coefficient changes and monic division used here work over arbitrary $\mathbb{C}$ -algebras, so the correspondence preserves local multiplicities.

If $\mathcal{X}_p$ is reduced, Proposition 5.6 gives S_p simple points of $\mathcal{Z}_p$, with $P_A(0) \neq 0$ at every point. Each quadratic in (6.5) then has two distinct roots. The nodal fiber has $2S_p = M_p$ simple points. In the finite coefficient family, the image of the closed locus where the matching Jacobian vanishes is closed. Its complement contains (6.4), and hence meets the nonsingular-lattice locus in a nonempty open set. Every fiber in that open set is reduced, proving the theorem. $\square$

The factor two in M_p retains the two choices of background in (2.5). That transformation preserves the evolution equation and the lattice; it changes the operator P and the speed. At the nodal fiber it exchanges the two roots in (6.5), which are distinct when $\mathcal{X}_p$ is reduced. In that case the two backgrounds are distinct on a generic lattice, and the M_p pairs represent $M_p/2$ elliptic waves, up to translations and Galilean shifts, each counted with both of its backgrounds. On special nonsingular lattices the two backgrounds may coalesce, as Section 9.2 shows.

6.1 Elliptic pairs of purely dispersive equations

Corollary 6.2. *Let p be even and assume (R_{p-1}) . On every fixed nonsingular lattice, the elliptic pairs whose operator is an even polynomial form a closed subscheme $\mathcal{E}_p^{\text{sym}}(g_2, g_3)$ of $\mathcal{E}_p(g_2, g_3)$, defined by $u_j = 0$ for odd j , and*

$$\text{length } \mathcal{E}_p^{\text{sym}}(g_2, g_3) = 2F_p. \quad (6.6)$$

If $\mathcal{X}_p$ is reduced, $\mathcal{E}_p^{\text{sym}}(g_2, g_3)$ is reduced for (g_2, g_3) in a nonempty Zariski-open set of nonsingular parameters.

Proof. An even operator has an even profile, by Corollary 3.2. Conversely, if the profile (6.1) is even, the odd part of (1.1) is $\sum_{j \text{ odd}} a_j D^j V = 0$; the functions $D^j V$ have distinct pole orders, so $a_j = 0$ for odd j . On the subspace $u_j = 0$, j odd, the residual of Step 1 is even, and there remain the equations and unknowns of even weight in (6.3):

$$\begin{array}{l|l} \text{variables} & 2, 4, \dots, p-2, p \\ \text{equations} & p+2, p+4, \dots, 2p-2, 2p. \end{array}$$

The omitted leading forms vanish identically on the subspace, so the remaining ones have only the trivial common zero, by Step 3. Weighted Bézout gives

$$\frac{(p+2)(p+4) \cdots (2p-2)(2p)}{2 \cdot 4 \cdots (p-2)p} = 2 \binom{p-1}{p/2-1} = 2F_p.$$

At the node, reflection is $Q \mapsto 1 - Q$. An even nodal profile is $\beta + v_A$ with $\iota A = A$, so the nodal fiber is (6.5) with $A \in \mathcal{X}_p^{\text{sym}}$. If $\mathcal{X}_p$ is reduced, it has $2F_p$ simple points, by Theorem 5.8 and Proposition 5.6, and Step 5 applies without change. $\square$

For odd p there is no fixed-lattice analogue. A purely dissipative equation has a profile that is $-P(0)$ plus a function odd about the pole, and its matching system has one equation more than it has unknowns. An exact computation on a sample lattice shows that for $p = 3, 5, 7$ it has no solution there, and hence none on a generic lattice. At $p = 3$ such pairs occur exactly when $g_2 = 0$, as (9.7) shows for $B = 0$.

7 Which orders are settled

Each formula needs two facts at a given p . The first is the rigidity condition (R_{p-1}) of Section 5. It gives N_p , M_p , F_p and $2F_p$ as counts with multiplicity: the multiplicities of the pairs add up to these numbers. The second is that every rational-exponential pair is simple, that is, $\mathcal{X}_p$ is reduced. Then the pairs are distinct, and the elliptic pairs are distinct on a generic lattice. On a special lattice several elliptic pairs can merge, while the count with multiplicity stays the same; Section 9.2 shows this at order three. Table 2 records what is known about each fact.

The prime-index argument excluded solutions at infinity by reducing the continuous convolution modulo ℓ . The same argument proves (5.10) whenever $2p-1 = \ell^e$ is an odd prime power (Appendix A). It gives the count with multiplicity, but it does not show that the pairs are distinct.

More generally, the exact arithmetic certificates supplied with the paper establish (R_d) for every $d \leq 72$, hence

$$\text{length } \mathcal{X}_p = \binom{2p-1}{p-1}, \quad \text{length } \mathcal{E}_p(g_2, g_3) = 2 \binom{2p-2}{p-2} \quad (2 \leq p \leq 73). \quad (7.1)$$

Table 2: What is known. For elliptic pairs, distinctness is asserted only on a generic lattice.

Statement	Known for
Counts with multiplicity	$2p - 1$ a prime or a prime power (proved); every $p \leq 73$ (exact certificate)
All N_p pairs distinct	$2p - 1$ prime (proved); $p = 5, 8, 11$ (exact computations); hence every $p \leq 12$
All F_p pairs of the pure subfamily distinct	whenever all N_p pairs are distinct; every $p \leq 14$ (exact computations)
Both statements, every p	conjectured

In the same range, Theorem 5.8 and Corollary 6.2 give length $\mathcal{X}_p^{\text{sym}} = F_p$ and, for even p , length $\mathcal{E}_p^{\text{sym}}(g_2, g_3) = 2F_p$. Together with the ancillary proof, the certificates also cover infinite families of higher orders. The certificates hold for all complex coefficients, not only real or rational ones. The ancillary proof and the full certificate are described in Appendix A.

Equation (7.1) is a statement about counts with multiplicity only. At composite indices, exact computations show that all pairs are simple at $p = 5$, $p = 8$ and $p = 11$ (Appendix A). With Theorem 5.4, all pairs are therefore distinct for every $p \leq 12$, and Theorem 6.1 gives M_p distinct elliptic pairs on a generic lattice at these orders. Inside the two pure subfamilies the computation is much smaller, and it shows that the F_p pairs are distinct for every $p \leq 14$, which includes the composite indices 9, 15, 21, 25 and 27.

8 What the count means for a single equation

The count lets the operator vary. We now read it for one evolution equation (2.1) with fixed complex coefficients. Recall that (2.1) is invariant under translations and under the Galilean shifts $u \mapsto u + k$, $x \mapsto x - kt$.

Proposition 8.1 (The waves of a fixed equation). *Let $\lambda \neq 0$. Up to a translation and a Galilean shift, the travelling waves of (2.1) that are rational in $e^{\lambda(x-ct)}$, with one pole modulo $(2\pi i/\lambda)\mathbb{Z}$, are*

$$u = b_p \lambda^p v_A(\lambda(x - ct)), \quad c = -b_p \lambda^p P_A(0), \quad (8.1)$$

where A ranges over the points of $\mathcal{X}_p$ that satisfy

$$\frac{b_j}{b_p} = \lambda^{p-j} [\rho^j] P_A, \quad 1 \leq j \leq p-1. \quad (8.2)$$

The data (A, λ) and $(\iota A, -\lambda)$ describe the same wave.

Proof. Such a wave has a finite limit as $e^{\lambda(x-ct)} \rightarrow 0$, by the argument of Section 4, and a Galilean shift makes this limit zero. Then $w_* = 0$ in (2.3), and (2.4) becomes (8.2) together with the formula for c . After a translation v has the normalization of Section 2, so $v = v_A$ and $P = P_A$ by Proposition 5.1. The last assertion follows from (5.20): the two waves differ by the constant $b_p \lambda^p s_p A(0)$, which is a Galilean shift. $\square$

Let $\pi(A) = ([\rho^{p-1}]P_A, \dots, [\rho^1]P_A)$ be the vector of lower coefficients of P_A , and let $\beta = (b_{p-1}/b_p, \dots, b_1/b_p)$ be the coefficient vector of the equation; this β is not the constant β of

Section 6. Condition (8.2) says that β is obtained from $\pi(A)$ by multiplying its k th entry by λ^k , for some $\lambda \neq 0$. As λ varies these vectors trace a curve, the orbit $\mathbb{C}^*\pi(A)$, and rescaling x moves an equation along it. This is the reason for fixing the rate: without it every pair would be a one-parameter family. Each of the N_p pairs marks one orbit, and several pairs may mark the same orbit.

Corollary 8.2 (Solvable equations). *Assume that $\mathcal{X}_p$ is finite, for instance that (R_{p-1}) holds.*

- (i) *Equation (2.1) has a travelling wave that is rational in an exponential, with one pole per period, if and only if β lies in one of the orbits $\mathbb{C}^*\pi(A)$, $A \in \mathcal{X}_p$. There are at most $(N_p + F_p)/2$ such orbits.*
- (ii) *If $\beta \neq 0$, there are finitely many such waves up to translations and Galilean shifts.*
- (iii) *If $p \geq 3$, the orbits form a set of dimension at most one in $\mathbb{C}^{p-1}$. A generic equation (2.1) has no such wave.*

Proof. Part (i) restates Proposition 8.1. For the bound, $\pi(\iota A) = (-1) \cdot \pi(A)$ by (5.20), so A and its mirror image ιA give the same orbit. The number of classes $\{A, \iota A\}$ is half the number of points plus half the number of fixed points, and these two numbers are at most length $\mathcal{X}_p \leq N_p$ and length $\mathcal{X}_p^{\text{sym}} \leq F_p$; the last inequality is the Bézout bound in the proof of Theorem 5.8, which does not require (R_{p-1}) . For (ii), if $\beta \neq 0$ and $\beta \in \mathbb{C}^*\pi(A)$, some coordinate $[\rho^j]P_A$ is nonzero, and (8.2) leaves at most $p - j$ values of λ . Part (iii) is a dimension count. $\square$

When $\beta = 0$ the equation is invariant under the scaling itself, and its waves come in one-parameter families of rates. Order two shows what the orbits mean. There are two of them, and both carry waves. The orbit $\beta = 0$ is the KdV equation, with its family of solitons of all rates. The orbit $\beta \neq 0$ consists of the equations with a nonzero Burgers coefficient. Each of them has one family of fronts, at the rate fixed by that coefficient. The table of Section 9.1 represents this front twice, by the mirror images $a = 1$ and $a = -1$; these are the data (A, λ) and $(\iota A, -\lambda)$ of Proposition 8.1. At $p = 3$, write $P = D^3 + BD^2 + \mu D + a_0$, so that $\beta = (B, \mu)$. The orbits are the curves $B^2 = \sigma\mu$ with the origin removed. For the ten pairs $A = \rho^2 + a\rho + b$ of Table 3, (9.5) gives $B = 4a$ and $\mu = a^2 + 19b$, hence the four classical values

$$\sigma = \frac{16a^2}{a^2 + 19b} \in \left\{0, 16, \frac{144}{47}, \frac{256}{73}\right\}. \quad (8.3)$$

Here the bound $(N_3 + F_3)/2 = 6$ is not attained, because the two pairs with $a = 0$ share $\sigma = 0$, and $(\pm 1, 0)$ and $(\pm i, 0)$ share $\sigma = 16$.

For odd p the classification turns Corollary 8.2 into a statement about all meromorphic travelling waves.

Corollary 8.3 (Generic equations with p odd). *Let $p \geq 5$ be odd and assume (R_{p-1}) . The coefficient vectors $\beta \in \mathbb{C}^{p-1}$ for which (2.1) has a nonconstant meromorphic travelling wave lie in an algebraic subset of dimension at most two. In particular a generic equation has none. For $p = 3$ this set is the union of the four curves $B^2 = \sigma\mu$ of (8.3).*

Proof. By Theorem 3.1, such a wave is rational, rational in an exponential, or elliptic, with one pole per period. Take $\lambda = 1$ in (2.3), so that $[\rho^j]P = b_j/b_p$ for $1 \leq j \leq p - 1$. The rational-exponential waves give a set of dimension at most one, by Corollary 8.2. For a rational wave (4.1), the argument of Step 3 in the proof of Theorem 6.1 applies verbatim, with the simple-pole term

$r_1 z^{-1}$ included: it gives $b = a_0 = 0$, and the Laplace correspondence gives $A \mid H_A$ for a monic A of degree $p - 1$, now without the restriction $A(0) = 0$. By (R_{p-1}) , $A = \rho^{p-1}$, hence $P = D^p$ and $\beta = 0$. For the elliptic waves, the family of all $\mathcal{E}_p(g_2, g_3)$ is finite over the (g_2, g_3) -plane, by Step 4 of the same proof. It has dimension two, so its image under $P \mapsto \beta$ has dimension at most two. For $p = 3$, Section 9.2 gives the elliptic condition $B^2 = 16\mu$, which is one of the four curves; the point $\beta = 0$ lies on all of them. $\square$

The same dimension count applies to the one-pole waves of every order $p \geq 3$ for which (R_{p-1}) holds: a generic equation has no travelling wave that is rational, rational in an exponential, or elliptic with one pole per period. For even p , however, Theorem 3.1 excludes other meromorphic waves only when the equation is purely dispersive.

9 Examples

The examples have one purpose: to show the counts of Table 1 at work, and to connect them with the classical lists. Orders two and three are small enough for the full family to be treated by hand (Sections 9.1 and 9.2); the second of them also shows what multiplicity means for elliptic pairs. At higher orders we list the pairs of the two pure subfamilies (Sections 9.3 and 9.4). The normalizations are those of Section 2.

9.1 Order two: KdV–Burgers

The integrated travelling-wave equation for

$$u_t + uu_x + \nu u_{xx} + \delta u_{xxx} = 0, \quad \delta \neq 0,$$

has order $p = 2$. Its exact front solutions have been studied in several equivalent forms [23, 29, 47]. The long-wave reductions in Kudryashov et al. [34] also contain this balance of dissipation and dispersion.

Here $A(\rho) = \rho + a$ has a single coefficient, and the matching system is one equation in one unknown. Direct polynomial division gives

$$\text{rem}_A S_A = -\frac{a(a-1)(a+1)}{6}, \quad P_A = \rho^2 + 5a\rho + a^2 - 6a - 1, \quad v_A = 12((a+1)Q - Q^2). \quad (9.1)$$

Thus there are exactly $N_2 = 3$ simple pairs:

a	$P_A(\rho)$	v_A
-1	$(\rho - 2)(\rho - 3)$	$-12Q^2$
0	$\rho^2 - 1$	$12Q(1 - Q) = 3 \operatorname{sech}^2(z/2)$
1	$(\rho - 1)(\rho + 6)$	$12Q(2 - Q).$

(9.2)

The pulse $a = 0$ is the KdV soliton. It is the single pair of the purely dispersive subfamily, $F_2 = 1$. The two fronts belong to mixed equations, and the reflection ι exchanges them: for a given equation with $\nu \neq 0$ they are one and the same front, read in the two directions (Proposition 8.1). At unit rate they are two different pairs, and the count $N_2 = 3$ includes both. In the original variables, (2.3) and (2.4) read

$$u(x, t) = u_* + \delta \kappa^2 v_A(\kappa(x - ct)), \quad \nu = \delta \kappa(5a), \quad u_* - c = \delta \kappa^2(a^2 - 6a - 1).$$

In particular, a nonzero ν fixes the front rate to $\kappa = \nu/(5a\delta)$, $a = \pm 1$, in agreement with the classical parameter restriction.

For a fixed lattice the elliptic candidate is $V = -12X + b$. Its residual is

$$-12a_1Y - 12(a_0 + b)X + 6g_2 + a_0b + \frac{1}{2}b^2.$$

Consequently the complete elliptic system is

$$a_1 = 0, \quad v = -12\wp - a_0, \quad a_0^2 = 12g_2. \quad (9.3)$$

There is no one-pole elliptic member with nonzero Burgers coefficient: both elliptic pairs belong to the KdV equation, in agreement with $M_2 = 2F_2 = 2$. They are the two backgrounds (2.5) of one cnoidal wave. For $g_2 \neq 0$ they are distinct and simple. For $g_2 = 0$, $g_3 \neq 0$, they coincide, and the lattice carries one pair of multiplicity two.

Since $2p - 1 = 3$ is prime, the counts at order two are unconditional. The classification of Theorem 3.1 applies to the KdV subfamily $a_1 = 0$. For the mixed operators, the Fisher example in Section 10 explains why the same conclusion cannot be expected.

9.2 Order three: Kuramoto–Sivashinsky

Let $p = 3$, $s_3 = 60$, and $A(\rho) = \rho^2 + a\rho + b$. The profile and monic operator are

$$\frac{v_A}{60} = 2Q^3 - (a + 3)Q^2 + (a + b + 1)Q, \quad (9.4)$$

$$P_A(\rho) = \rho^3 + 4a\rho^2 + (a^2 + 19b)\rho - a^3 + 7ab - 5a - 30b. \quad (9.5)$$

The remainder of S_A is $(E_1\rho + E_0)/30$, where

$$E_0 = ab(a^2 - 7b + 5), \quad E_1 = a^4 - 8a^2b + 11b^2 + 10b - 1. \quad (9.6)$$

The system $E_0 = E_1 = 0$ is solved by cases on the factors of E_0 . If $b = 0$, then $a^4 = 1$. If $a = 0$, then $(b + 1)(11b - 1) = 0$. In the remaining case $a^2 = 7b - 5$, and the second equation becomes $4(b - 2)(b - 3) = 0$. This exhausts the system and gives Table 3. The Jacobian determinant is nonzero at every point, so all $N_3 = 10$ pairs are simple, as Theorem 5.4 predicts for $2p - 1 = 5$.

Table 3: All ten rational-exponential pairs at order three. The determinant is taken in the order (E_0, E_1) against (a, b) . The last column describes v_A on the real axis, at unit rate.

a	b	Number	$\det \partial(E_0, E_1)/\partial(a, b)$	Real profiles
± 1	0	2	-24	Pulses
$\pm i$	0	2	-16	Not real
0	-1	1	144	Front
0	1/11	1	576/121	Front
± 3	2	2	-144	Fronts
± 4	3	2	384	Fronts

There are ten complex pairs, eight with real coefficients. All eight are smooth and bounded on the real axis: two pulses and six fronts. For example,

$$A = (\rho - 1)(\rho - 2), \quad v_A = 120Q^3, \quad P_A = (\rho - 3)(\rho - 4)(\rho - 5).$$

These are the classical solutions in hyperbolic functions [26, 27, 39]. The complete parameter list is tabulated by Kudryashov and Zargaryan [33, Table 1] and Conte and Musette [5, Table 1]. Their tables contain six parameter cases because they identify the two signs of the dispersion coefficient. These six cases are the orbits of the reflection ι , which is $(a, b) \mapsto (-a, b)$ here: four orbits of two pairs, and the two fixed pairs with $a = 0$. Exhaustion of this classical list follows from Eremenko [19].

The two pairs with $a = 0$ are the $F_3 = 2$ pairs of the purely dissipative subfamily, the Kuramoto–Sivashinsky equation without dispersion. Both are fronts, as Theorem 5.8 requires; the pulses $(\pm 1, 0)$ need dispersion. By Corollary 8.2, an equation with $P = D^3 + BD^2 + \mu D + a_0$ has a wave of this kind exactly when $B^2 = \sigma\mu$ for one of the four values (8.3). The pairs $(\pm 1, 0)$ and $(\pm i, 0)$ belong to the same equations, with $\sigma = 16$; their rates differ by a factor i , so for a real equation the second wave is periodic in x , and its real form has poles on the real axis.

Elliptic pairs, and the lattices on which they merge. An elliptic solution must have zero residue at its pole. For $P = D^3 + BD^2 + \mu D + a_0$ the Laurent residue is $60\mu/19 - 15B^2/76$, so $B^2 = 16\mu$. With this restriction the entire elliptic matching scheme is

$$\begin{aligned} P &= D^3 + BD^2 + \frac{B^2}{16}D + a_0, \\ v &= -60\wp' - 15B\wp - \frac{B^3}{64} - a_0, \\ B^4 &= 3072g_2, \quad a_0^2 = \frac{13B^6}{2048} - 2160g_3. \end{aligned} \tag{9.7}$$

Read with the operator fixed, this is the classical solution of Kudryashov [27]: $w = v + a_0$ solves the usual integrated equation $w''' + Bw'' + \mu w' + w^2/2 + K = 0$ with $K = -a_0^2/2$. Necessity of the restriction $B^2 = 16\mu$ is proved by Hone [21], and the classification of all meromorphic KS solutions is completed by Eremenko [19]. For the count we read (9.7) the other way: fix g_2, g_3 and solve for B and a_0 .

For a typical lattice, the equation for B has four solutions, and the equation for a_0 then gives two choices for each of them: the two backgrounds $\pm a_0$ of (2.5). These are the $M_3 = 8$ pairs. When the lattice changes, some of the pairs can meet. The multiplicity of the resulting point records how many nearby pairs have come together, so the count with multiplicity can remain eight even when fewer than eight distinct pairs remain.

Pairs meet exactly where the Jacobian of the last two equations of (9.7) vanishes, and that Jacobian is a nonzero constant times B^3a_0 . Let

$$j(\Lambda) = \frac{1728g_2^3}{g_2^3 - 27g_3^2}.$$

If $B = 0$, then $g_2 = 0$, hence $j = 0$. If $a_0 = 0$ and the lattice is nonsingular, elimination gives

$$675g_3^2 = 169g_2^3, \quad j = -300.$$

These conditions are sufficient as well. Thus:

$j(\Lambda)$	distinct pairs	total with multiplicity
$j \notin \{0, -300\}$	8	8×1
-300	6	$2 \times 2 + 4 \times 1$
0	2	$2 \times 4.$

(9.8)

The two exceptional cases show the two ways in which pairs can meet.

At $j = -300$, the four values of B remain distinct. For two of them, the two choices of a_0 meet at zero: the two backgrounds of one wave coincide. Each produces one pair of multiplicity two, while the other four pairs remain distinct. There are six distinct pairs, with total multiplicity $2 + 2 + 1 + 1 + 1 + 1 = 8$. For instance, $g_2 = 1/12$, $g_3 = 13/1080$ gives the double points $(B, a_0) = (\pm 4, 0)$.

At $j = 0$, all four values of B meet at $B = 0$. The two roots of $a_0^2 = -2160g_3$ remain distinct, since $g_3 \neq 0$, so there are two pairs, each of multiplicity four. Both have the operator $D^3 + a_0$, which is purely dissipative, and the total is again $4 + 4 = 8$.

The count stays eight for an exact reason. The coefficients eliminated on the way to (9.7) are determined linearly, with nonzero rational pivots, so its last two equations carry the full matching system, multiplicities included. The first has four roots B and, for each of them, the second has two roots a_0 , counted with multiplicity. This gives eight solutions on every lattice. This is the form in which Theorem 6.1 holds on an arbitrary lattice, and (9.8) is the exceptional set of that theorem at order three.

9.3 Purely dispersive equations: Kawahara and seventh-order KdV

Order four. For the Kawahara equation [15, 24], the integrated, shifted and monic travelling-wave operator is

$$P = D^4 + bD^2 + a_0. \quad (9.9)$$

The classification for this equation, using its first integral, is already established by Demina and Kudryashov [15]; see also Ye et al. [48]. Their first integral is the order-four instance of (3.7).

By Theorem 5.8, the pairs with an even operator are those with $\iota A = A$, that is, $A = \rho^3 + c\rho$. All of them are pulses: the profile is even about its pole, $v(Q) = v(1 - Q)$. The quotient and remainder are

$$\begin{aligned} P_A &= \rho^4 + 13c\rho^2 + \frac{31c^2 - 70c + 7}{3}, \\ v_A &= 280(1 + c)Q(1 - Q) - 1680Q^2(1 - Q)^2, \\ \text{rem}_A S_A &= -\frac{\rho}{420}(c + 1)(31c^2 - 31c + 10). \end{aligned} \quad (9.10)$$

The three distinct roots are

$$c = -1, \quad c = \frac{1}{2} \pm \frac{3i}{2\sqrt{31}}.$$

They give $F_4 = \binom{3}{1} = 3$ simple pairs. These are the three simply periodic parameter choices of Demina and Kudryashov [15, Section 3], in the present normalization: the parameter in their Eq. (31) equals $1/c$ here. At $c = -1$,

$$v = -105 \operatorname{sech}^4(z/2), \quad P = (D^2 - 4)(D^2 - 9). \quad (9.11)$$

This is the familiar solitary wave of the Kawahara equation; the other two pairs are complex at unit real rate. The unrestricted order-four count is $N_4 = 35$.

All elliptic members of (9.9) have the form

$$v = -1680\wp^2 - \frac{280b}{13}\wp + 168g_2 + \frac{31b^2}{507} - a_0. \quad (9.12)$$

Substitution gives exactly two conditions:

$$F(b; g_2, g_3) := 31b^3 - 42588bg_2 - 4745520g_3 = 0, \quad (9.13)$$

$$a_0^2 = 23184g_2^2 + \frac{7980}{169}b^2g_2 + \frac{101520}{13}bg_3. \quad (9.14)$$

These are the Weierstrass profiles and parameter relations obtained from Laurent series by Demina and Kudryashov [15, Section 3], expressed in the normalization (1.1). In their notation, $b = -\beta$, $a_0^2 = C_0^2 + 12C_1$; the profile change is $v = -6(w - w_*)$, where w_* is a constant equilibrium and $a_0 = C_0 - 6w_*$. The first equation is cubic in b , and the second is monic quadratic in a_0 , so there are $3 \times 2 = 6$ solutions, counted with multiplicity, on every fixed lattice: three values of b , each with its two backgrounds. This is the number $2F_4$ of Corollary 6.2. At $(g_2, g_3) = (1, 0)$, the cubic has three distinct roots and the right side of (9.14) is nonzero at all of them, so the six pairs are simple on a nonempty open set of lattices. The unrestricted elliptic count at this order is $M_4 = 30$; elliptic formulas with nonzero odd derivative coefficients in the full fifth-order evolution equation are given by Kudryashov and Sinelshchikov [37].

Order six. The seventh-order KdV equation has the operator $P = D^6 + a_4D^4 + a_2D^2 + a_0$; its meromorphic travelling waves are studied by Dang [10]. Here $A = \rho^5 + c_1\rho^3 + c_2\rho$ and $s_6 = 5544$. The remainder of S_A has two coefficients, at ρ^3 and at ρ , which are polynomials in c_1, c_2 of weights eight and ten; the ancillary script prints them. Eliminating c_2 , which is determined by c_1 at every solution, gives

$$(c_1 + 5)(c_1 + 10)(49c_1^2 - 5c_1 + 10)g_6(c_1) = 0, \quad (9.15)$$

where g_6 is an irreducible sextic without real roots, also printed by the script. The ten roots are distinct, in agreement with $F_6 = \binom{5}{2} = 10$. Two pairs are real:

A	P_A	v_A
$\rho(\rho^2 - 1)(\rho^2 - 4)$	$(\rho^2 - 9)(\rho^2 - 16)(\rho^2 - 25)$	$10395 \operatorname{sech}^6(z/2)$
$\rho(\rho^2 - 1)(\rho^2 - 9)$	$(\rho^2 - 4)(\rho^2 - 25)(\rho^2 - 71)$	$10395 \operatorname{sech}^4(z/2)(1 + \operatorname{sech}^2(z/2))$

(9.16)

All ten pairs are pulses. An exact computation on a sample lattice gives $2F_6 = 20$ for the elliptic pairs of this family.

9.4 Purely dissipative equations of order five

At $p = 5$ the purely dissipative operators are $P = D^5 + a_3D^3 + a_1D + a_0$, and $A = \rho^4 + c_1\rho^2 + c_2$. All 126 pairs at $p = 5$ are simple (Section 7), so there are exactly $F_5 = 6$ distinct pairs here. They are

$$A = (\rho^2 - 1)(\rho^2 - 4), \quad A = (\rho^2 - 1)(\rho^2 - 9), \quad A = (\rho^2 - 1)\left(\rho^2 + \frac{1}{5}\right), \quad (9.17)$$

and the three pairs with

$$1105c_1^3 - 2969c_1^2 + 5570c_1 - 4000 = 0, \quad c_2 = \frac{5c_1^3 + 21c_1 - 10}{79c_1 + 210}.$$

The cubic has one real root. All six pairs are fronts, as Theorem 5.8 requires, and the four with real coefficients are, after the scales are restored, the solutions (1.4) of Kudryashov and Migita [32]. For instance, $A = (\rho^2 - 1)(\rho^2 - 4)$ gives

$$v_A = 5040Q^3(10 - 15Q + 6Q^2), \quad P_A = \rho^5 - 55\rho^3 + 1254\rho - 2520,$$

a front from 0 to $5040 = -2P_A(0)$.

10 Open questions and the boundary of the classification

Two questions remain open in the enumeration, and one limitation of the classification deserves a precise statement.

Rigidity. The first question is the continuous-convolution rigidity assertion

$$0 \neq A \in \mathbb{C}[\rho], \quad A \mid \int_0^\rho A(t)A(\rho-t) dt \implies A = c\rho^{\deg A}.$$

Through the polynomial Laplace correspondence used in the proof of Theorem 6.1, it says that, after fixing the pole at zero, any rational one-pole solution of (1.1) vanishing at infinity would have to be $c_* z^{-p}$, with operator $P = D^p$. Algebraic and ℓ -adic arguments prove it in the range $\deg A \leq 72$, corresponding to $2 \leq p \leq 73$, and in infinite arithmetic families of higher degrees. Rigidity in arbitrary degree remains a conjecture. By Theorems 5.3 and 6.1, it would establish both enumeration formulas with algebraic multiplicities at every order.

Simplicity. The second question is whether every point of $\mathcal{X}_p$, the rational-exponential matching system, is simple. The elliptic simplicity assertion is generic in the lattice and is derived from this one in Theorem 6.1. At a solution $S_A = AC_A$, the tangent map of the system is

$$J_A(h) = \text{rem}_A(2(A \star h) - C_A h), \quad \deg h < d. \quad (10.1)$$

The question is whether this map always has zero kernel. The prime-index argument proves this after reduction modulo ℓ , where the map becomes multiplication by a polynomial coprime to $\bar{A}$; a uniform replacement is missing at composite indices. One might expect simplicity to follow from the uniqueness of the Laurent expansion in Theorem 3.1. It does not: Laurent uniqueness concerns a fixed operator, whereas (10.1) allows the operator to vary with the profile.

Conjecture 10.1. Continuous-convolution rigidity holds in every degree, and the matching Jacobian (10.1) is invertible at every point of $\mathcal{X}_p$. Consequently $\#\mathcal{X}_p(\mathbb{C}) = \text{length } \mathcal{X}_p = \binom{2p-1}{p-1}$ for every $p \geq 2$.

The conjecture would also imply M_p simple elliptic pairs on a generic fixed lattice at every order, F_p distinct pairs in the two pure subfamilies, and that the rational-exponential waves of every purely dissipative equation are fronts (Theorem 5.8). Simplicity on every lattice is false already at order two; (9.8) describes the full exceptional set at order three.

Mixed equations with p even. The hypothesis of Theorem 3.1 is a separate issue. For even p it requires a purely dispersive equation, and it cannot simply be dropped. The known example comes from the Fisher equation. Its travelling waves of nonzero speed satisfy (1.1) with $p = 2$ and a mixed operator, which is also the profile equation of KdV–Burgers. At the special speed of Ablowitz and Zeppetella [1] it has a one-parameter family of solutions, given in Kudryashov [30], Conte et al. [6, Lemma 4.4] and McCue et al. [41]; see also Kudryashov [28]. In our normalization it is

$$v(z) = -12\lambda^2 e^{2\lambda z} \wp(e^{\lambda z} - 1; 0, g_3), \quad P(D) = D^2 - 5\lambda D + 6\lambda^2, \quad \lambda \neq 0.$$

Substitution follows from $(D^2 - 5\lambda D + 6\lambda^2)[t^2 U(t)] = \lambda^2 t^4 U''(t)$, $t = e^{\lambda z}$. For $g_3 = 0$ and $\lambda = 1$ this is the front in the first row of (9.2). For generic $g_3 \neq 0$ it is globally meromorphic and lies outside $\mathcal{W}$. Further examples of meromorphic solutions outside $\mathcal{W}$ are discussed by Ng and Wu

[43, Remark 1.11]. Thus a classification for mixed operators with p even must allow a larger class of functions than $\mathcal{W}$.

For mixed operators of even degree $p \geq 4$ we do not know whether meromorphic solutions outside $\mathcal{W}$ occur. Methods for constructing elliptic and rational-exponential solutions without the finiteness property are developed by Demina [14].

A Exact verification and ancillary material

This appendix records one lemma used in the proof of Theorem 5.4 and describes the computer checks that accompany the paper.

The lemma transfers the absence of solutions at infinity from characteristic ℓ to characteristic zero. Let homogeneous polynomials $F_1, \dots, F_m$, with coefficients in $\mathbb{Z}_{(\ell)}$, have no nonzero common zero over $\overline{\mathbb{F}}_\ell$. Then they have no nonzero common zero over $\mathbb{C}$. Indeed a complex projective zero yields a zero over $\overline{\mathbb{Q}}$ after fixing one nonzero coordinate. Choose a place above ℓ , extend it if necessary, and scale all coordinates by the inverse of a coordinate of smallest valuation. The coordinates are integral and one is a unit. Their reductions give a nonzero common zero over $\overline{\mathbb{F}}_\ell$, contradicting the hypothesis. The argument needs the algebraic closure of the residue field; a search only over $\mathbb{F}_\ell$ would not suffice.

Prime powers. Let $2p-1 = \ell^e$ be an odd prime power. For $\deg A = (\ell^e - 1)/2$, the normalized continuous convolution satisfies

$$\ell^e H_A(\rho) \equiv u\rho^{\ell^e} \pmod{\ell}, \quad u \neq 0.$$

Every bilinear coefficient other than the leading one has denominator valuation at most $e-1$, whereas the leading coefficient has valuation $-e$. This follows either from the factorial formula or from

$$\frac{i!j!}{(i+j+1)!} = \sum_{r=0}^j \frac{(-1)^r \binom{j}{r}}{i+r+1}.$$

The lemma above then gives (R_{p-1}) , as in Step 3 of the proof of Theorem 5.4. This argument controls only the leading system, so it gives no simple-root lifting statement for the discrete matching system.

The ancillary material contains five kinds of checks.

1. `verify_manuscript.py` reconstructs the discrete and continuous convolutions, the low-order Laurent coefficients, and the Weierstrass differential algebra. It also checks the leading-term comparison used in Proposition 5.2. It checks the displayed order-two, order-three and order-four formulas by exact substitution, including the correspondence with the KS tables of Conte and Musette [5], Kudryashov and Zargaryan [33], the KS collision multiplicities, the Kawahara coefficient restriction and its literature normalization, the nodal correspondence, and the Fisher identity. These finite substitutions check the displayed formulas; the uniform proofs are the arguments in the text.
2. `p5_count.sing` is a self-contained rational SINGULAR calculation for the full order-five matching ideal. Its quotient dimension is 126. Adjoining the matching Jacobian determinant gives the unit ideal, so all 126 points are simple. The input is reconstructed from the original polynomial sum by the Python checker; the saved transcript states the two exact outputs.

3. At $p = 8$ the same two outputs, 6435 and the unit ideal, are obtained over the finite field $\mathbb{F}_{32003}$. Every point modulo 32003 is therefore simple and lifts uniquely by Hensel’s lemma, as in Step 4 of the proof of Theorem 5.4, and the lifts are all the points because the length is 6435. At $p = 11$ each of the 352716 points is certified over the 19-adic numbers, by Hensel’s lemma at the simple points of the reduction modulo 19 and by exact Newton-polygon and Newton–Kantorovich certificates at the singular ones; the archived certificate, with its description and the record of its validation, is in the repository, directory `certificates/`.
4. `rigidity/` supplies the proof and exact certificate for (R_d) , $d \leq 72$. For $2d + 1 = \ell^e + g$, $g \in \{0, 2, 4, 6\}$, the proof reduces the normalized continuous convolution to a residual system of degree at most g . The certificate retains every residual degree, with integer ideal-membership identities and certified prime factors. The replay reconstructs the inputs, multiplies out every identity, and checks all 72 arithmetic choices. It does not infer characteristic-zero nonexistence from a list of residue-field points.
5. `symmetry_checks/` contains the exact checks for the two pure subfamilies and for single equations: the identities of Proposition 5.7 with symbolic coefficients for $d \leq 5$; the lengths F_p for $p \leq 8$, and finite-field certificates, of the same kind as at $p = 8$, that the F_p pairs are distinct for every $p \leq 14$; the lengths $2F_p$ of Corollary 6.2 for $p \leq 6$ on a sample lattice; the six pairs (9.17) and the ten pairs (9.15), with the sextic factor in full; and the scaling (8.1) by substitution in the evolution equation.

A Lean 4 formalization accompanies the paper at <https://github.com/migita/binomial-count-meromorphic-waves>, directory `lean/`, with Lean and Mathlib pinned to version 4.33.1. It proves (R_d) for $2d + 1$ prime (`rigidity_prime_index`) and the count and simplicity statements of Theorem 5.4, both for the matching system (5.6) (`primeIndexCountAndSimplicity`) and for the normalized pairs (P, v) as solutions of (1.1) in the field of rational functions of e^z (`primeIndexRationalExponentialCountAndSimplicity`), together with the correspondence between the two formulations for every $p \geq 2$. The discrete convolution is defined there by the Bernoulli-polynomial power sums, and the remainder equations and their Jacobian are those of Section 5. The proof route is outlined in Remark 5.5 and documented in the repository. The files `EnumerationCheck.lean` and `Check.lean` verify that the proofs have exactly the stated propositions as their types and that the theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`; the project contains no `sorry`, added axiom or `native_decide`.

The scripts use exact rational or finite-field arithmetic. Instructions, dependency versions and checksums are included with the source package. The prime-index theorem does not depend on the computer calculations.

Tool and computational resource disclosure

This paper was written with substantial use of generative AI, which is disclosed here in the form recommended by the Leiden Declaration on Artificial Intelligence and Mathematics¹ and required by arXiv. No automated system is an author; credit and responsibility for the results, and for the attribution of prior work, rest with the author.

¹<https://leidendeclaration.ai/>, 2 June 2026.

- **gpt-6-astra** (OpenAI), August–September 2026. The main mathematical ideas of the paper, the proofs of Theorems 3.1, 5.3, 5.4 and 6.1, the rigidity condition (R_d) and its certificate, the exact verification scripts, the SINGULAR input, the first complete draft of the text, and the Lean 4 formalization of Remark 5.5 were produced by this model in dialogue with the author.
- OpenAI Codex v0.153.4, September 2026. Independent mathematical audit of the final revision, integration of the edits, and rebuilding and validation of the source package.
- Claude Code (Anthropic, Claude Fable 5.1), September 2026. The literature search behind the attributions in Section 1, the readability revision of all sections, and the leading-coefficient argument of Proposition 5.2, which Codex then checked independently. In the final revision it proposed and proved the results on the two pure subfamilies and on single equations (Corollary 3.2, Section 5.4, Corollary 6.2 and Section 8), with their exact checks, and reorganized the manuscript around them. It also audited the Lean formalization independently: the formal statements against the manuscript, a rebuild from source, and the axiom audit.

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